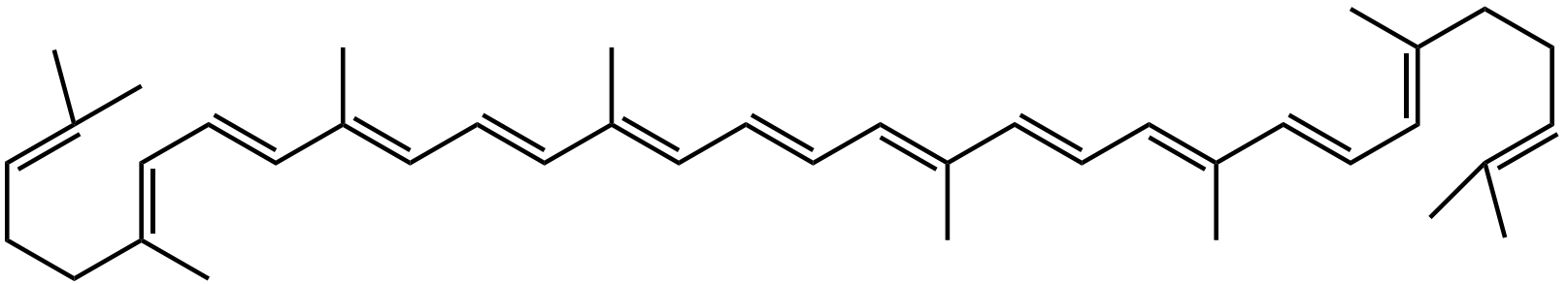
DELOCALIZATION AND CONJUGATION

- **Interaction between orbitals over many bonds**
- **Structure of aromatic compounds**
- **Representing one aspect of structure by curly arrows**
- **Stabilization by the sharing of electrons over more than two atoms**
- **Molecular shape and structure determine reactivity**
- **Where colour comes from**

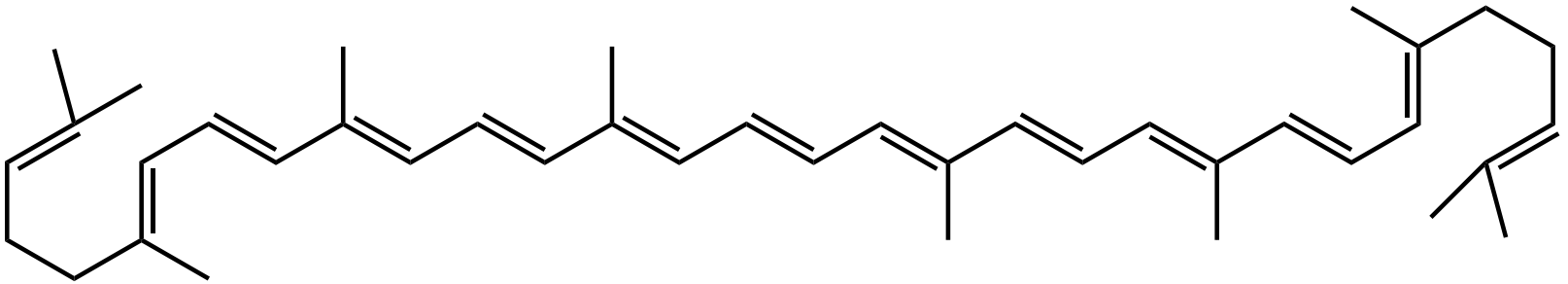
Conjugation: Some examples



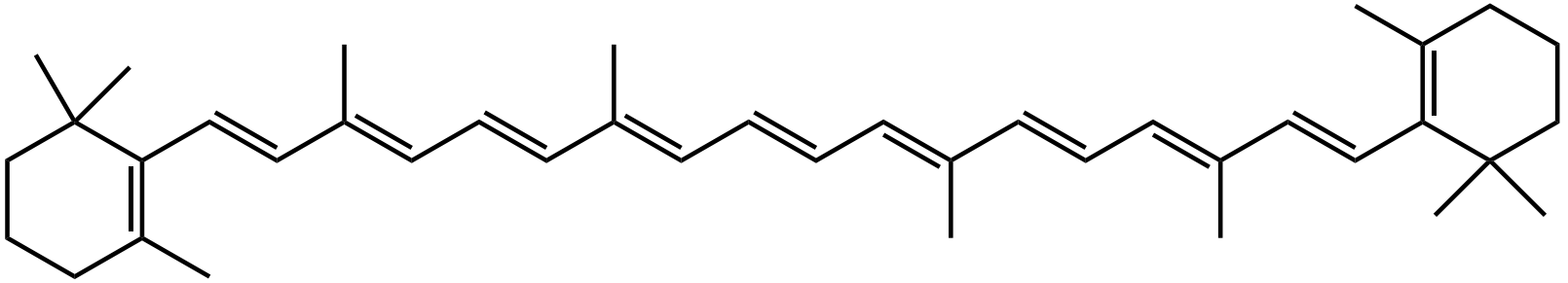
Lycopene, the red pigment in tomatoes, rose hips, and other berries

http://lpi.oregonstate.edu/infocenter/phytochemicals/carotenoids/

Conjugation: Some examples



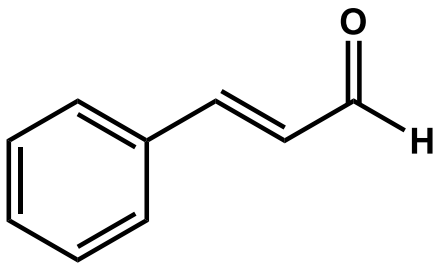
Lycopene, the red pigment in tomatoes, rose hips, and other berries



β-carotene, the red pigment in carrots and other vegetables

http://lpi.oregonstate.edu/infocenter/phytochemicals/carotenoids/

Conjugation: Some examples

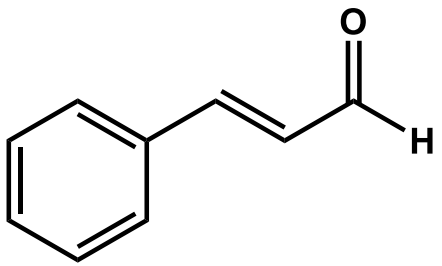


cinnamaldehehyde

Found in cinnamon



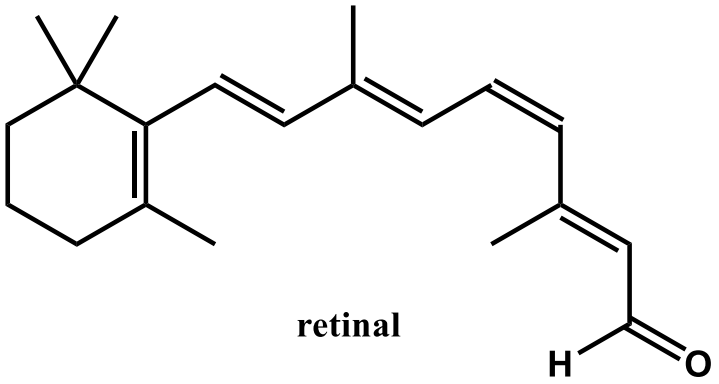
Conjugation: Some examples



cinnamaldehehyde

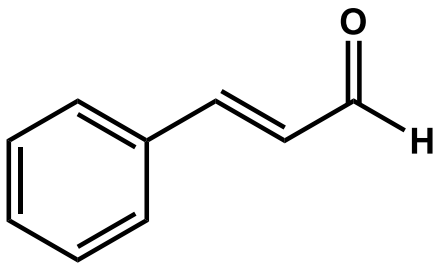
Found in cinnamon





retinal

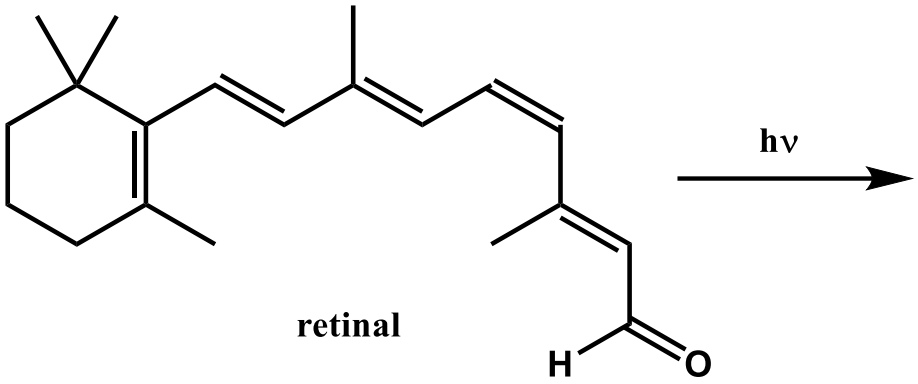
Conjugation: Some examples



cinnamaldehehyde

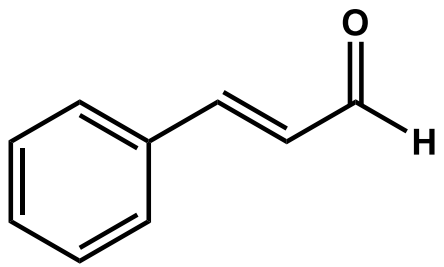
Found in cinnamon





Photoisomerization of retinal is the first step in vision

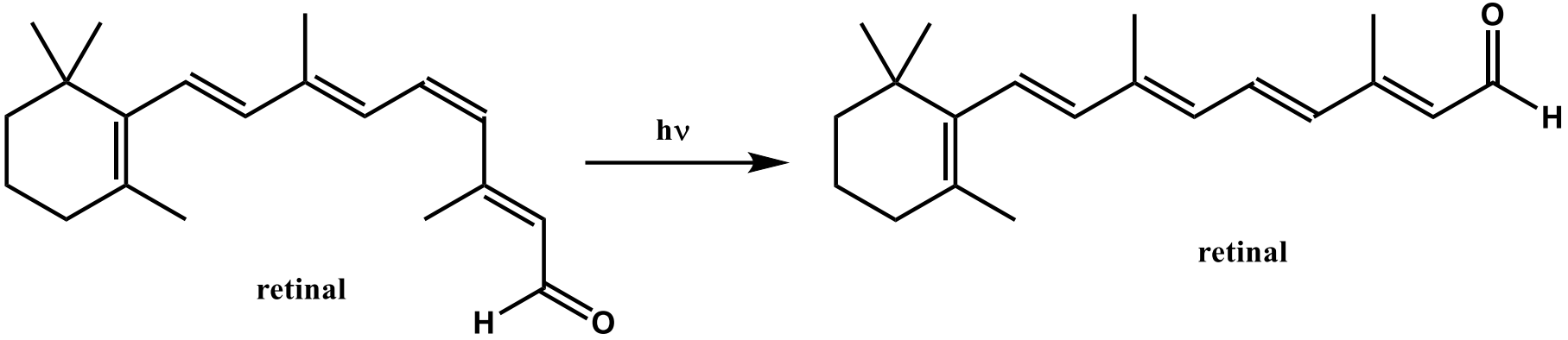
Conjugation: Some examples



cinnamaldehehyde

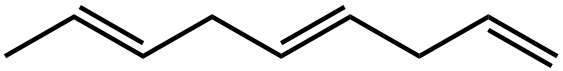
Found in cinnamon



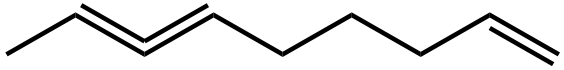


Photoisomerization of retinal is the first step in vision

Conjugation: Some examples

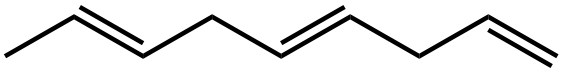


Isolated double bonds



Cumulated double bonds

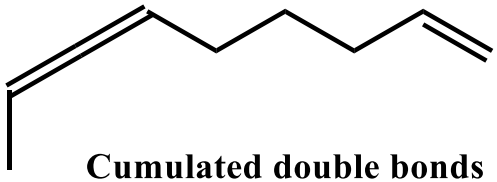
Conjugation: Some examples



Isolated double bonds



Cumulated double bonds



Cumulated double bonds



Conjugated double bonds

What are delocalization and conjugation?

- **Electrons are delocalized over the whole of a conjugated system. We use conjugation to describe bonds and delocalization to describe electrons.**

- **Conjugated is a system of p-orbitals linked together by alternating single and double bonds that connect into a single π system of delocalized electrons into adjacent p orbitals**

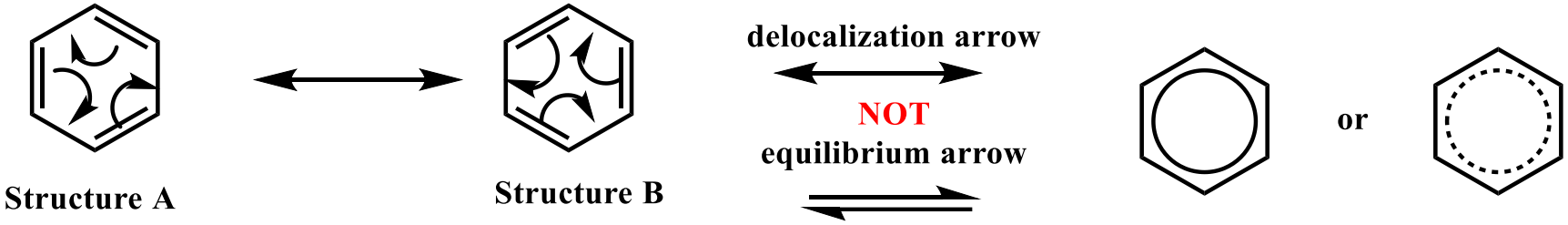
- **Conjugation focuses on the sequence of alternating double and single bonds while delocalization focuses on the molecular orbitals covering the whole system**

Delocalized π systems and aromaticity

- **EXPERIMENTAL FACT: Benzene has 3 double bonds but does NOT undergo addition reactions. WHY?**

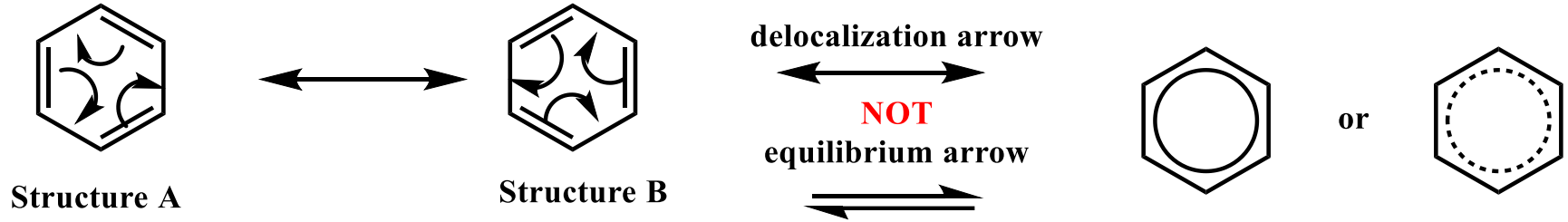
Delocalized π systems and aromaticity

- **EXPERIMENTAL FACT: Benzene has 3 double bonds but does NOT undergo addition reactions. WHY?**

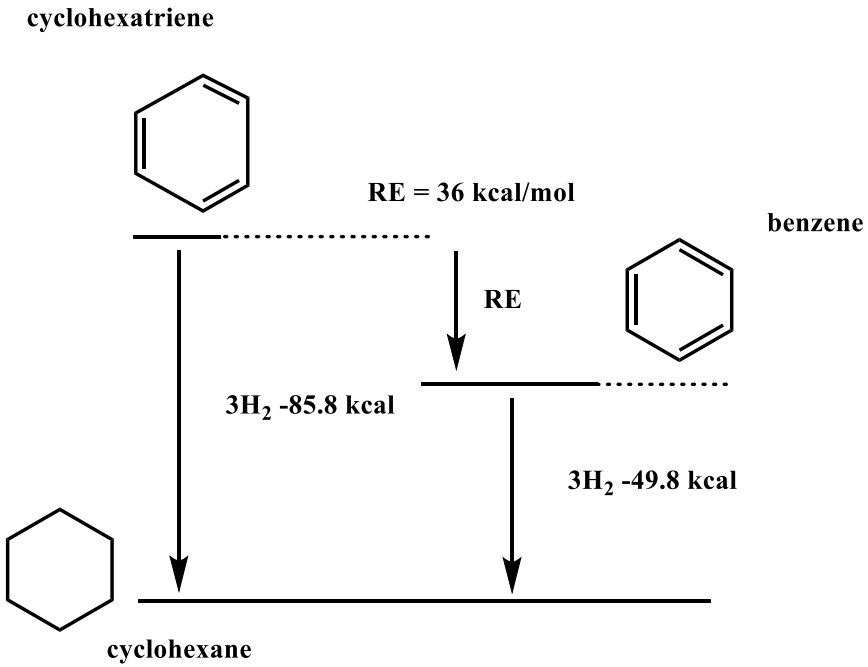


Delocalized π systems and aromaticity

- **EXPERIMENTAL FACT: Benzene has 3 double bonds but does NOT undergo addition reactions. WHY?**

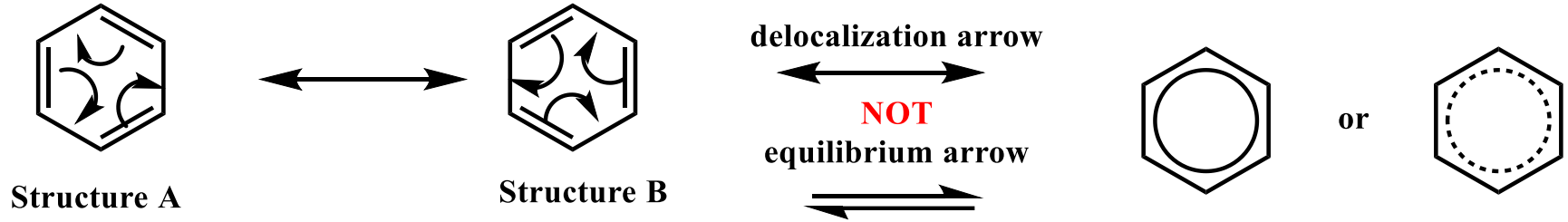


- **Compounds containing 3 or more adjacent overlapping p orbitals are thermodynamically stabilized by the extended overlap**

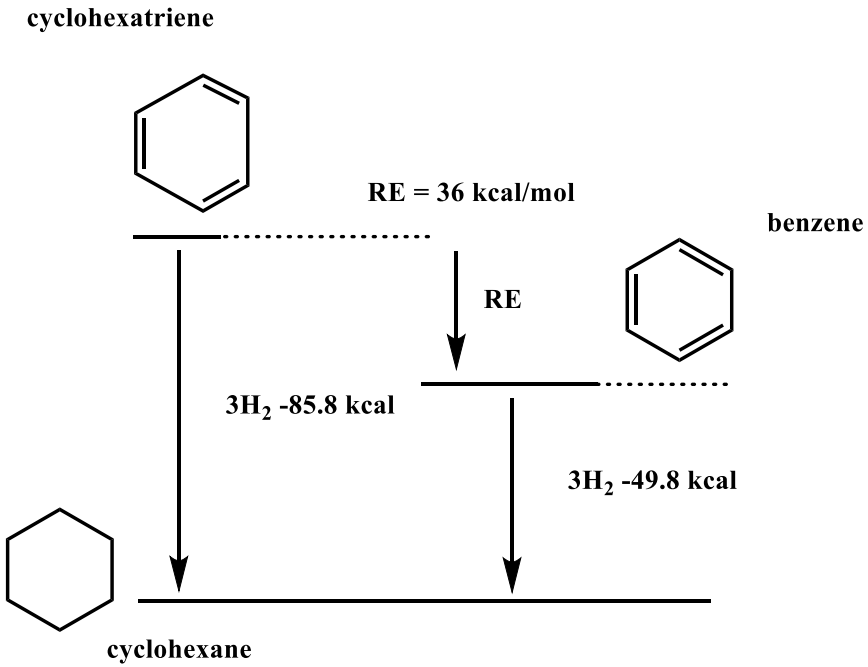


Delocalized π systems and aromaticity

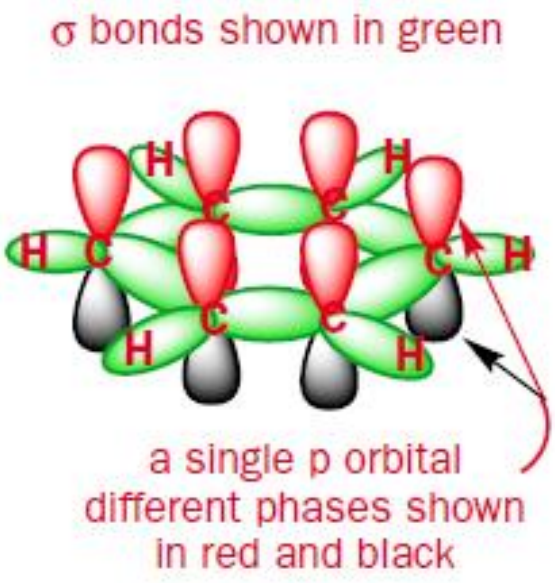
- **EXPERIMENTAL FACT: Benzene has 3 double bonds but does NOT undergo addition reactions. WHY?**



- **Compounds containing 3 or more adjacent overlapping p orbitals are thermodynamically stabilized by the extended overlap**
- **Delocalization arises whenever there are 2 or more structures for the same molecule that have the same arrangement of atoms but a different arrangement of electrons**

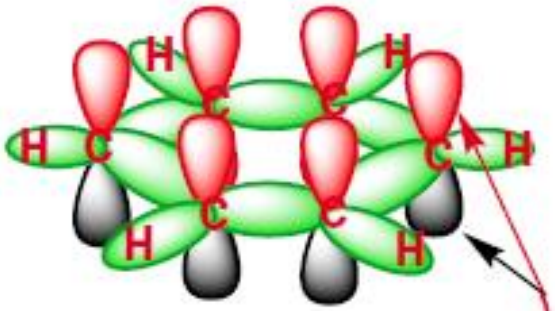


Benzene structure



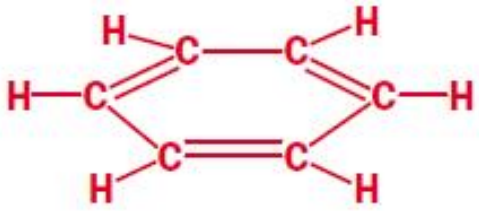
Benzene structure

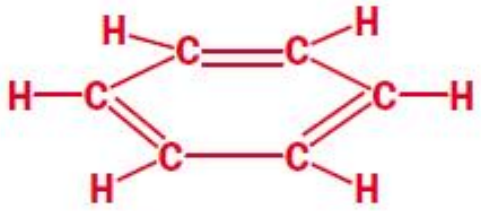
σ bonds shown in green

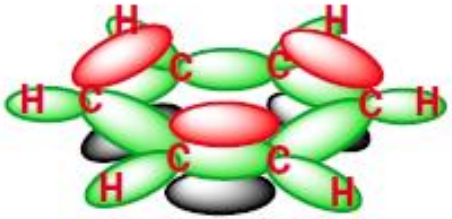


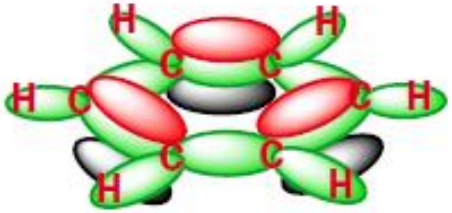
a single p orbital
different phases shown
in red and black

combining different pairs of p orbitals
puts the double bonds in different postions



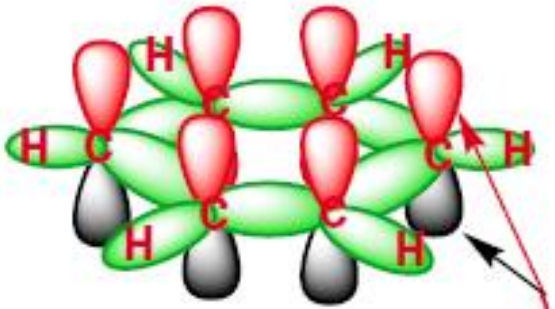






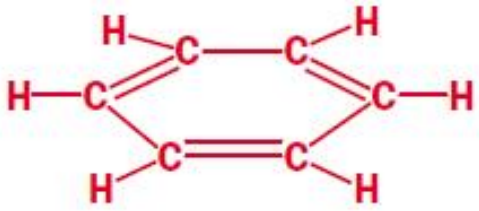
Benzene structure

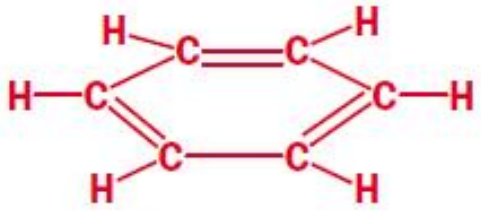
σ bonds shown in green

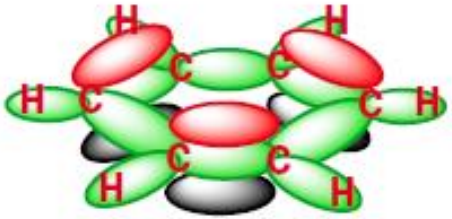


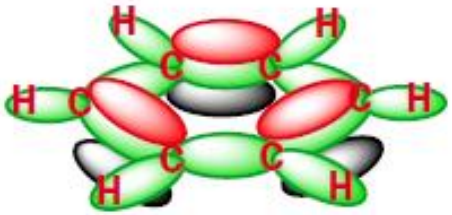
a single p orbital
different phases shown
in red and black

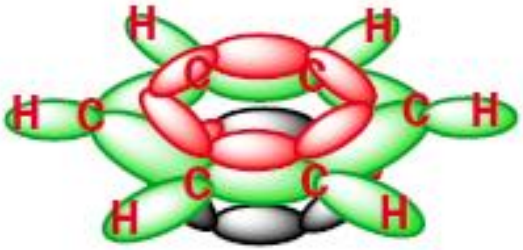
combining different pairs of p orbitals
puts the double bonds in different postions













the circle represents the
delocalized system

A hybrid

Remember the unicorn analogy







A unicorn is a mythical creature which is NOT sometimes a horse and sometimes a rhinoceros. It is a HYBRID of both

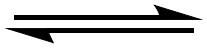


delocalization arrow

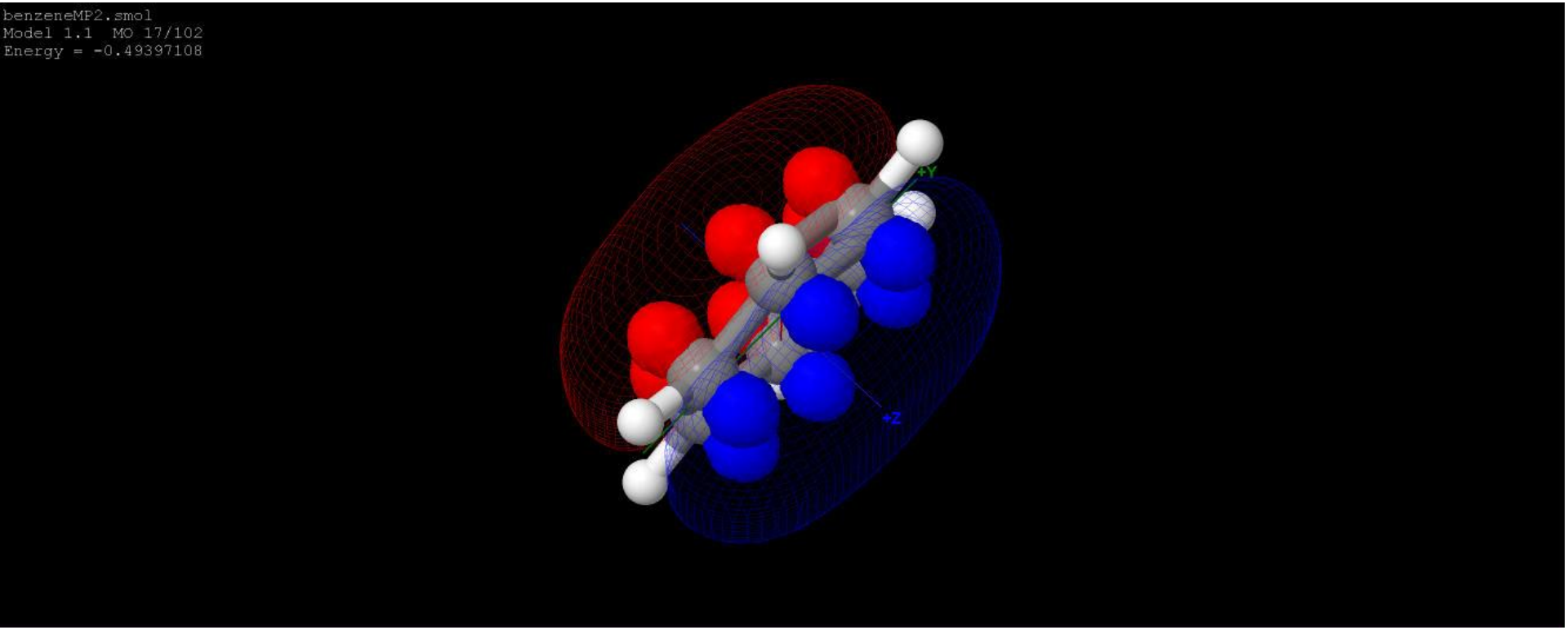


NOT

equilibrium arrow

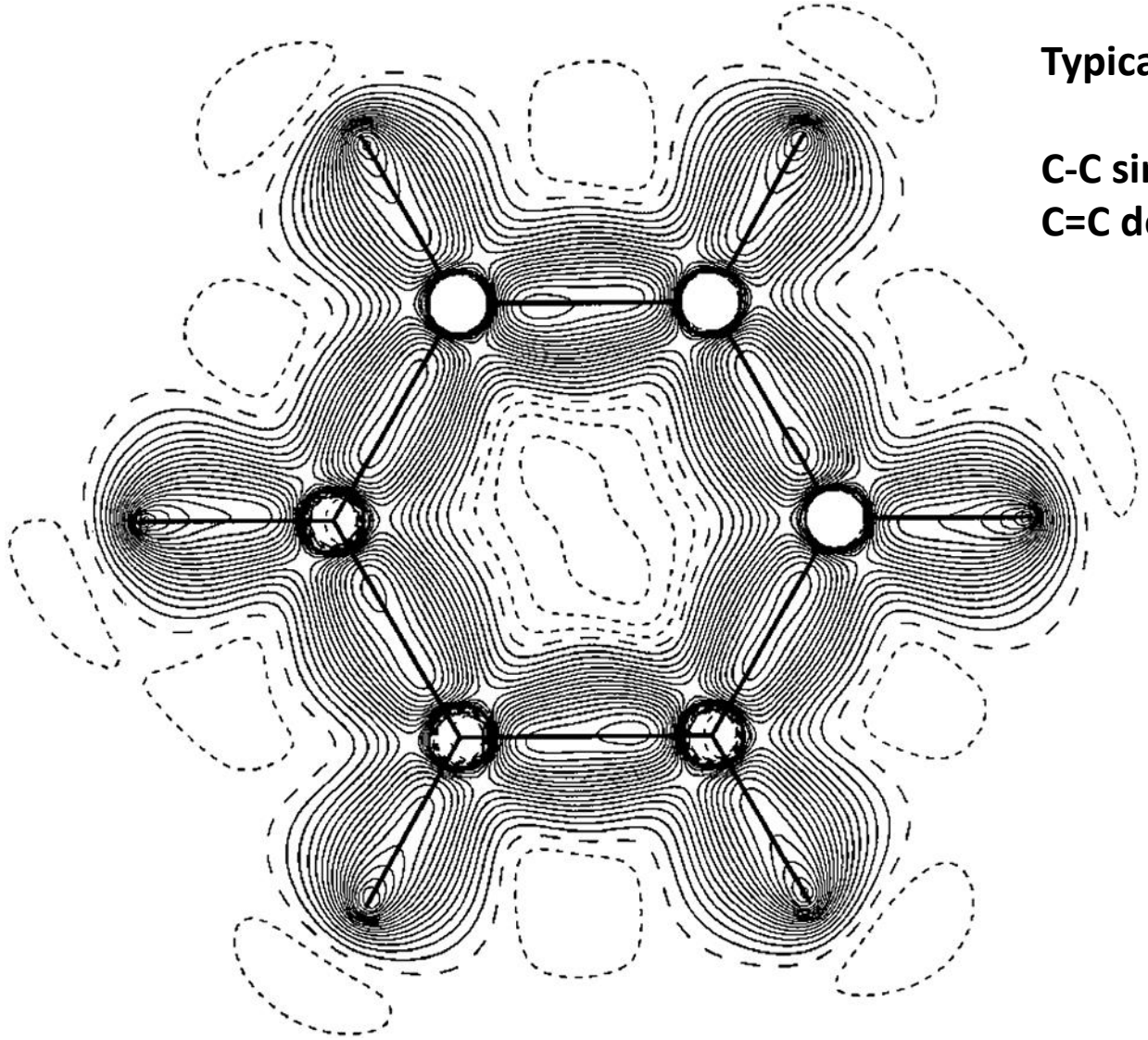


Benzene orbitals



Visit: http://www.chemtube3d.com

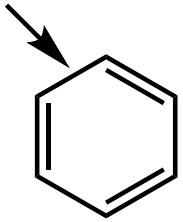
Benzene electron diffraction image



Typical values:

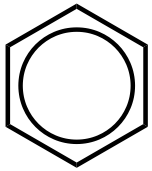
C-C single bond 154 pm (1.54Å)
C=C double bond: 134 pm (1.34Å)

All C-C bonds 139.5 pm (1.395Å)

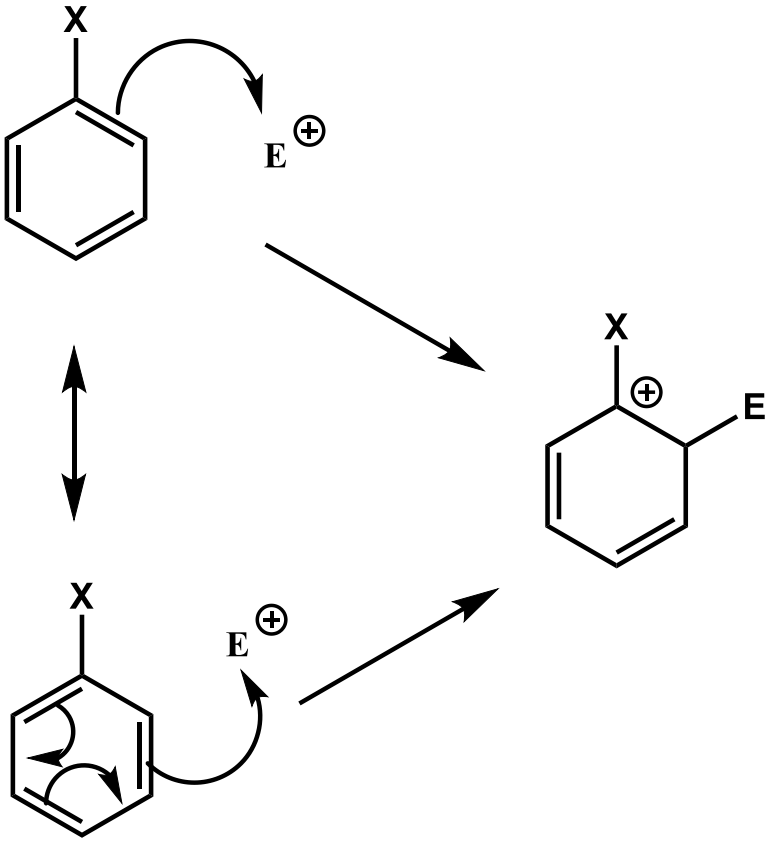


1pm= 10^{-12} meter= 0.01Å

Representation of benzene in mechanisms

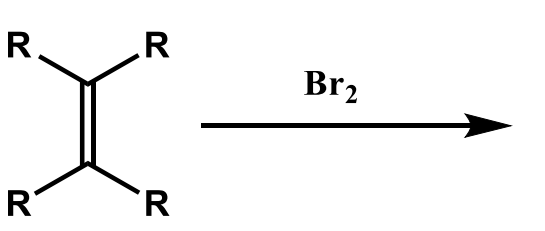


This representation cannot be used for mechanisms

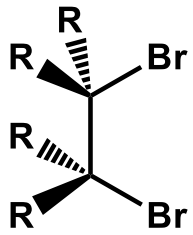


Regardless of the way you draw the benzene ring- both drawing give the same result but one uses more arrows to get there

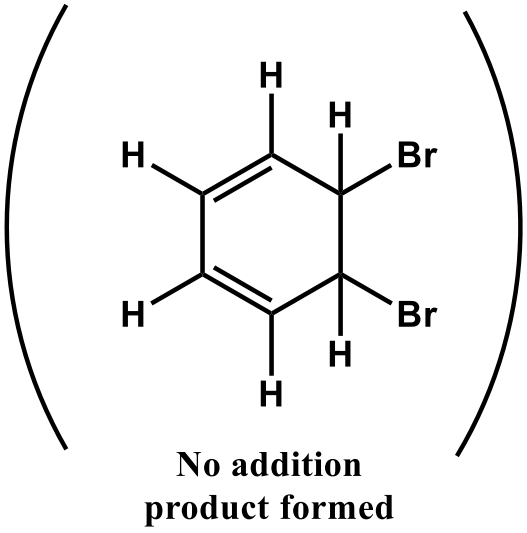
Reactivity of Benzene



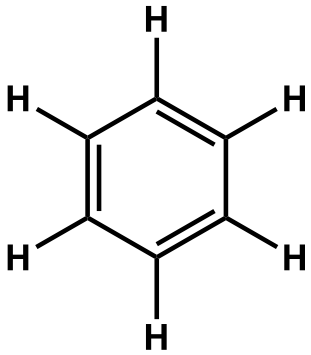
Br_2



BUT

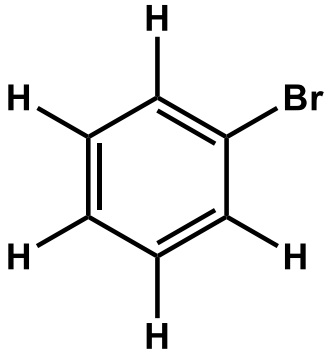


No addition product formed



Br_2

Fe catalyst



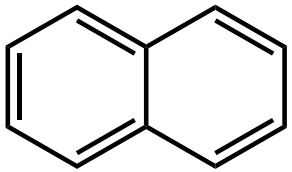
Hückel's Rule-Predictor of Aromaticity

A planar, cyclic molecule will be aromatic if it has $4n+2\pi$ electrons (n=0,1,2…)

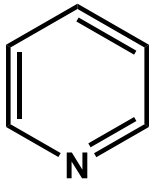
To be aromatic a compound must:

- **Be cyclic**
- **Follow Hückel's rule (have 2, 6 , 10, 14 … p electrons)**
- **Be able to be planar**
- **Have every atom in the ring able to participate in delocalizing the electrons by having p-orbital or a lone pair of electrons**

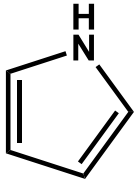
Examples of aromatic and non aromatic compounds



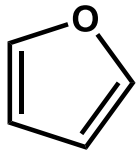
naphtalene



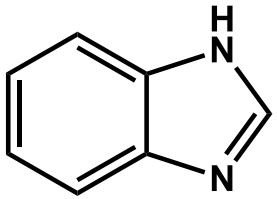
pyridine



pyrrole

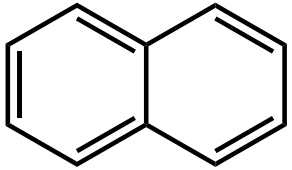


furan

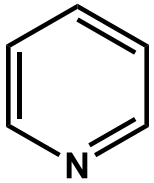


benzimidazole

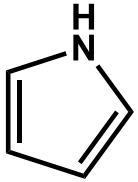
Examples of aromatic and non aromatic compounds



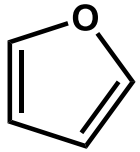
naphtalene



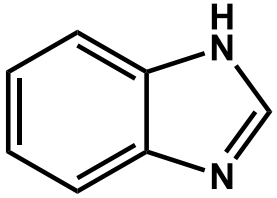
pyridine



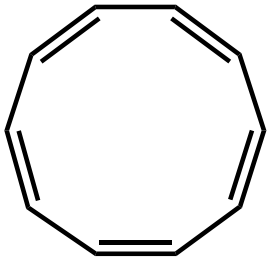
pyrrole



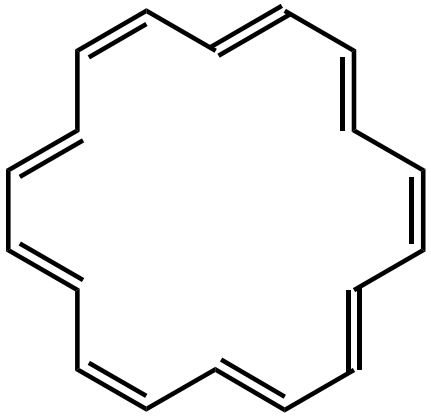
furan



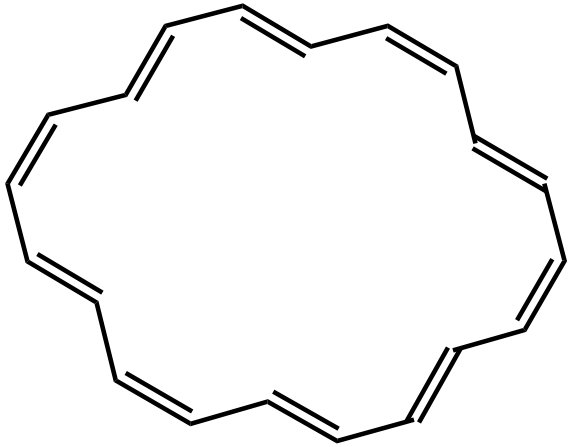
benzimidazole



all-cis-[10]-annulene



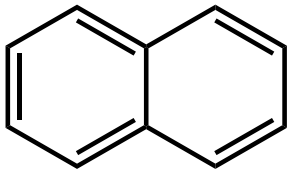
[18]-annulene



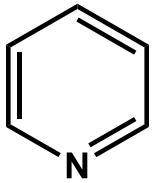
[20]-annulene

[n]-Annulenes: Conjugated monocyclic hydrocarbons

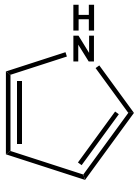
Examples of aromatic and non aromatic compounds



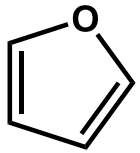
naphtalene



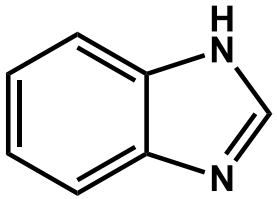
pyridine



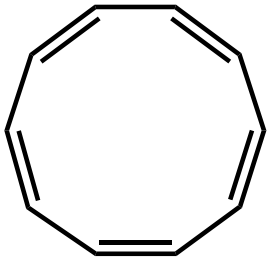
pyrrole



furan

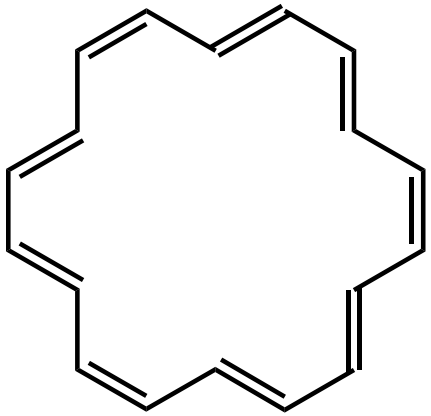


benzimidazole

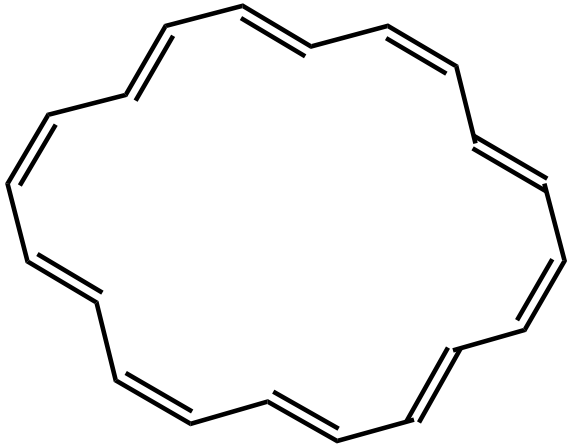


all-cis-[10]-annulene

obeys the (4n+2) rule but is not aromatic because it is too strained when planar



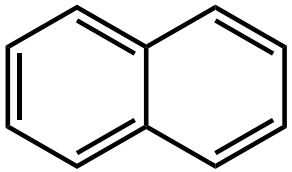
[18]-annulene



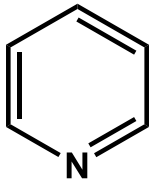
[20]-annulene

[n]-Annulenes: Conjugated monocyclic hydrocarbons

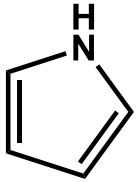
Examples of aromatic and non aromatic compounds



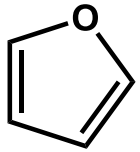
naphtalene



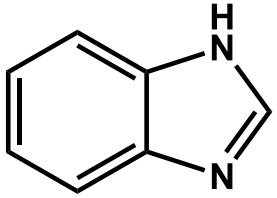
pyridine



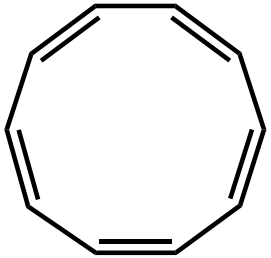
pyrrole



furan

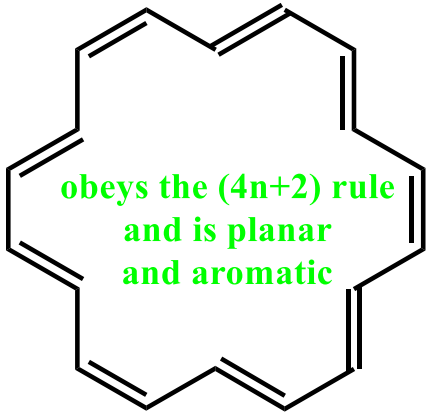


benzimidazole



all-cis-[10]-annulene

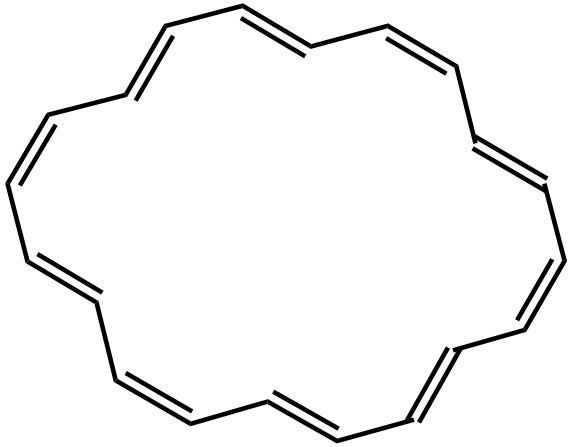
obeys the (4n+2) rule but is not aromatic because it is too strained when planar



obeys the (4n+2) rule and is planar and aromatic

[18]-annulene

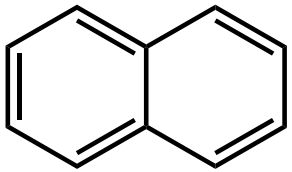
note the trans-trans-cis double bonds: all bond angles can be 120 ^{o}



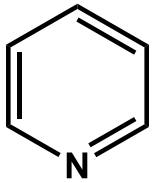
[20]-annulene

[n]-Annulenes: Conjugated monocyclic hydrocarbons

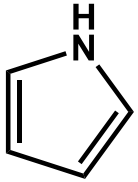
Examples of aromatic and non aromatic compounds



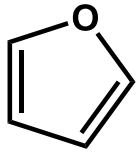
naphtalene



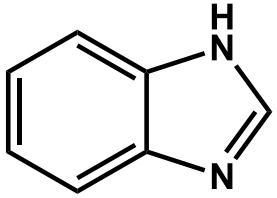
pyridine



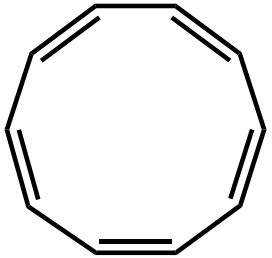
pyrrole



furan

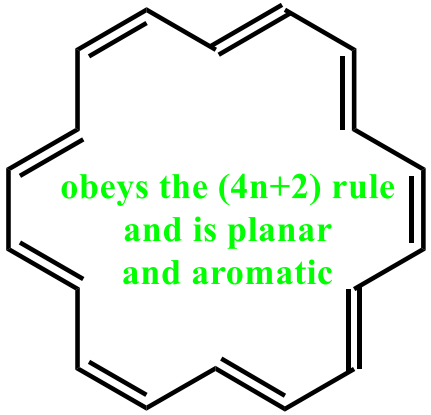


benzimidazole



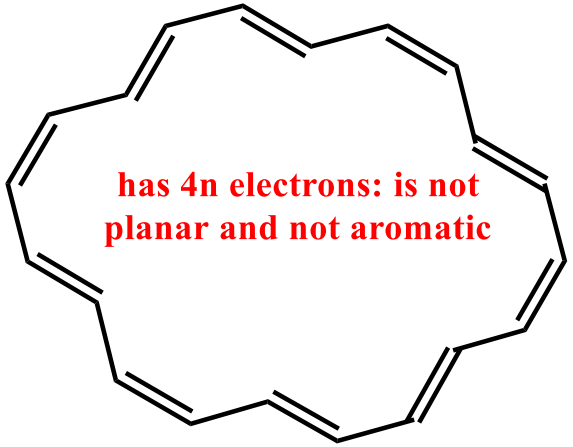
all-cis-[10]-annulene

obeys the (4n+2) rule but is not aromatic because it is too strained when planar



[18]-annulene

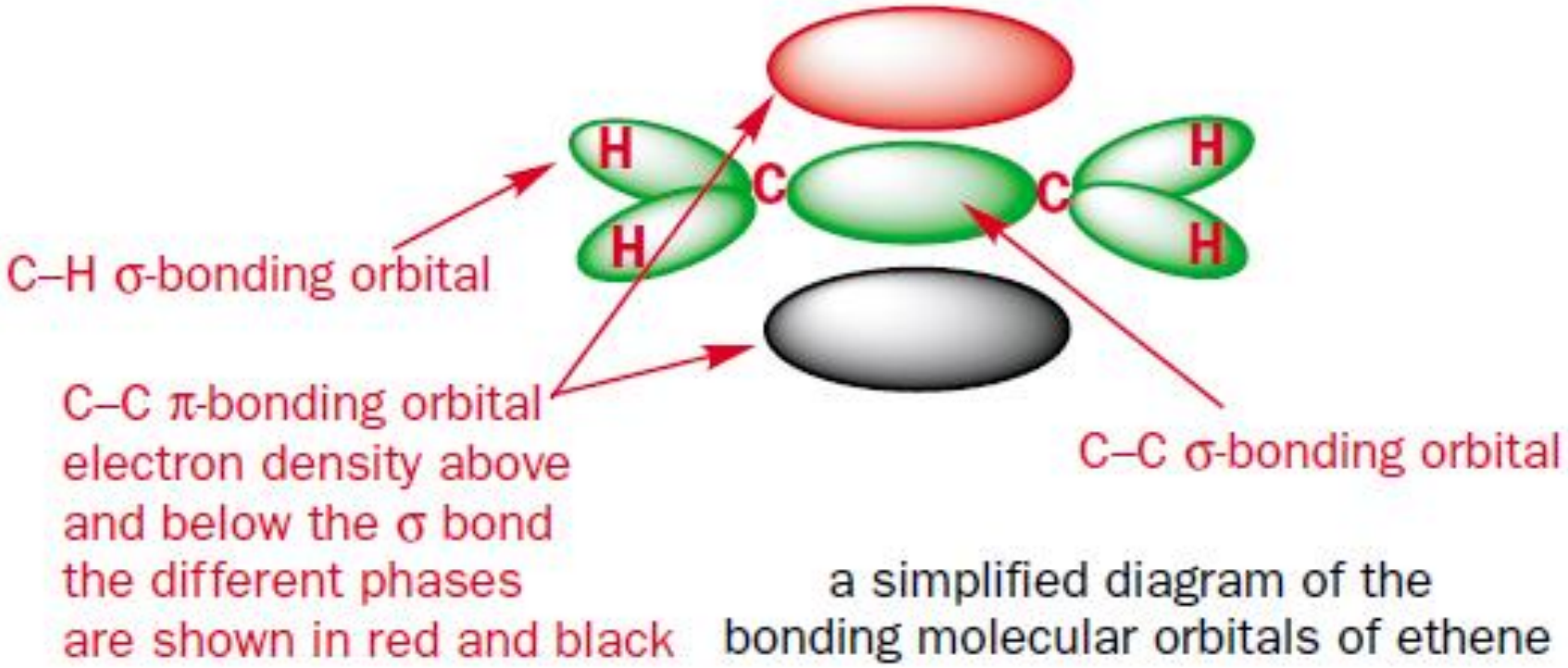
note the trans-trans-cis double bonds: all bond angles can be 120 ^{o}



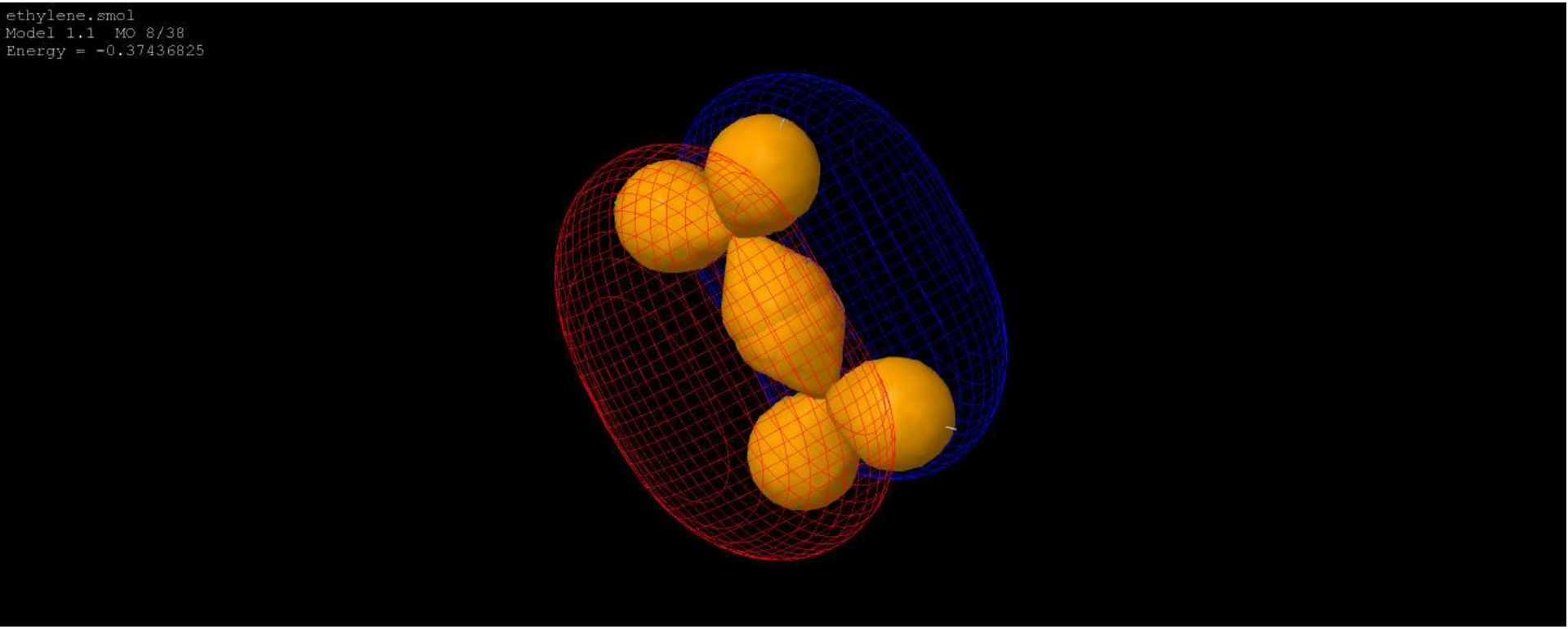
[20]-annulene

[n]-Annulenes: Conjugated monocyclic hydrocarbons

Ethene (Ethylene) structure

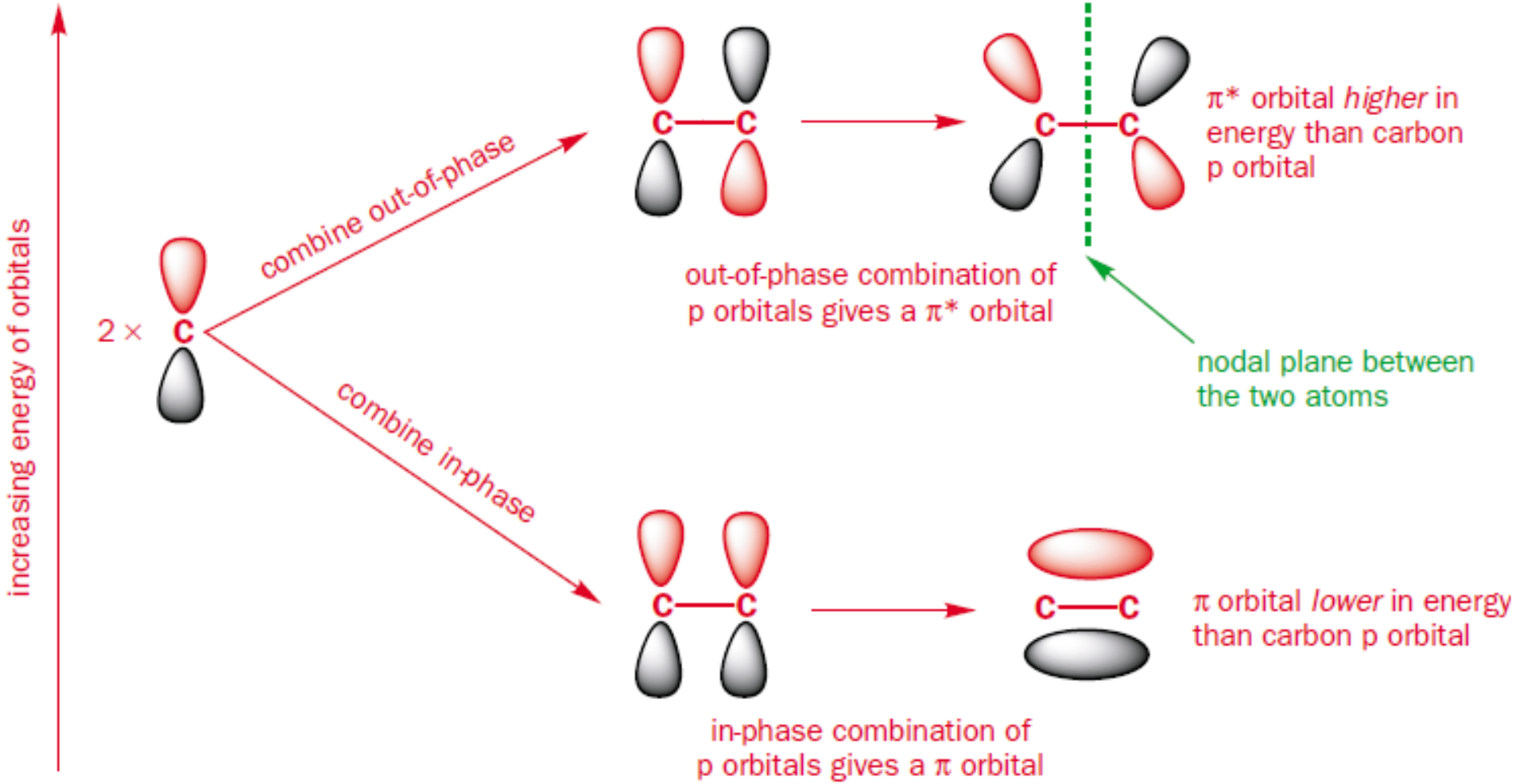


Ethene orbitals

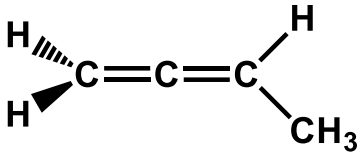


Visit: http://www.chemtube3d.com

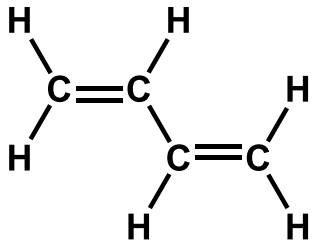
Ethene orbitals



Butadiene structure-The conjugation of 2 π bonds



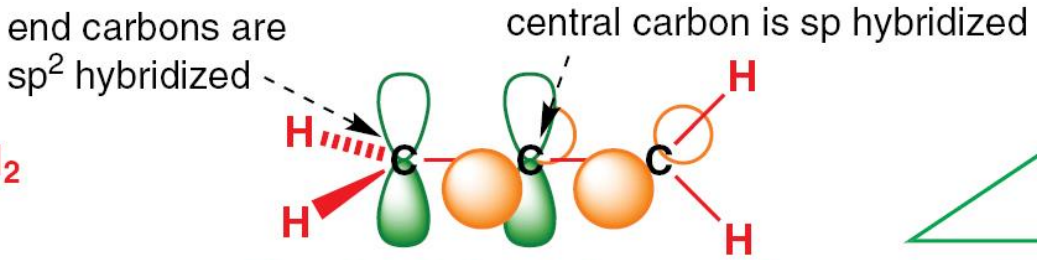
1,2-butadiene
an allene



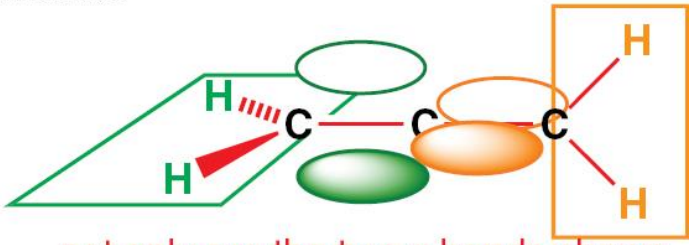
1,3-butadiene
a conjugated diene

$H_2C=C=CH_2$

allene

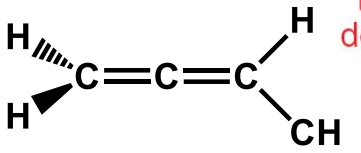


the π bonds formed as a result of the overlap of the individual p orbitals (shown here) must be at right angles to each other

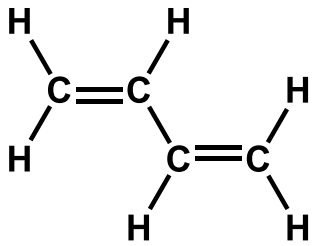


not only are the two π bonds shown in this diagram perpendicular, but the two CH_2 groups are too

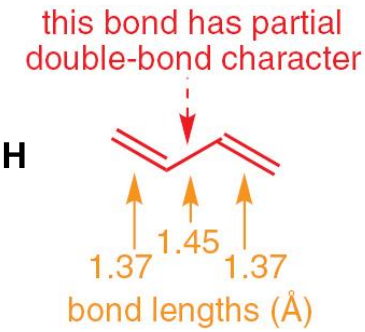
Butadiene structure-The conjugation of 2 π bonds



1,2-butadiene
an allene



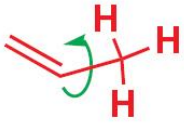
1,3-butadiene
a conjugated diene



typical values:
single bond: 1.54 Å
double bond: 1.34 Å



it takes 30 kJ mol^{-1} to rotate about this bond



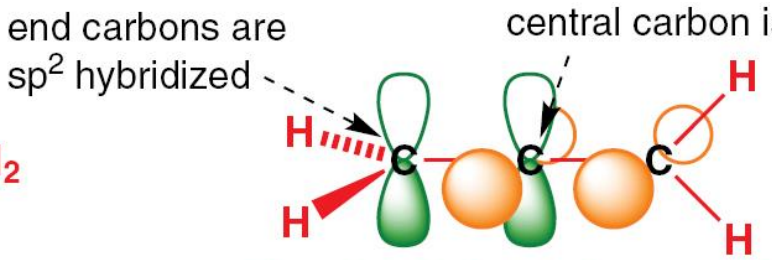
it takes 3 kJ mol^{-1} to rotate about this single bond



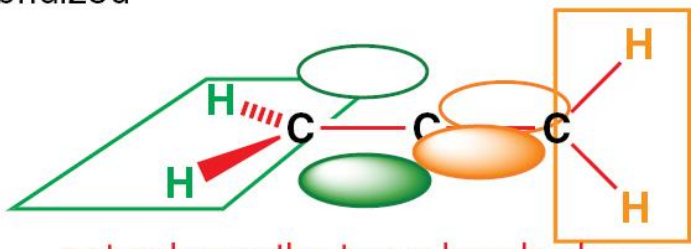
rotation about a full double bond needs more than 260 kJ mol^{-1}: it's essentially impossible

$H_2C{=}C{=}CH_2$

allene

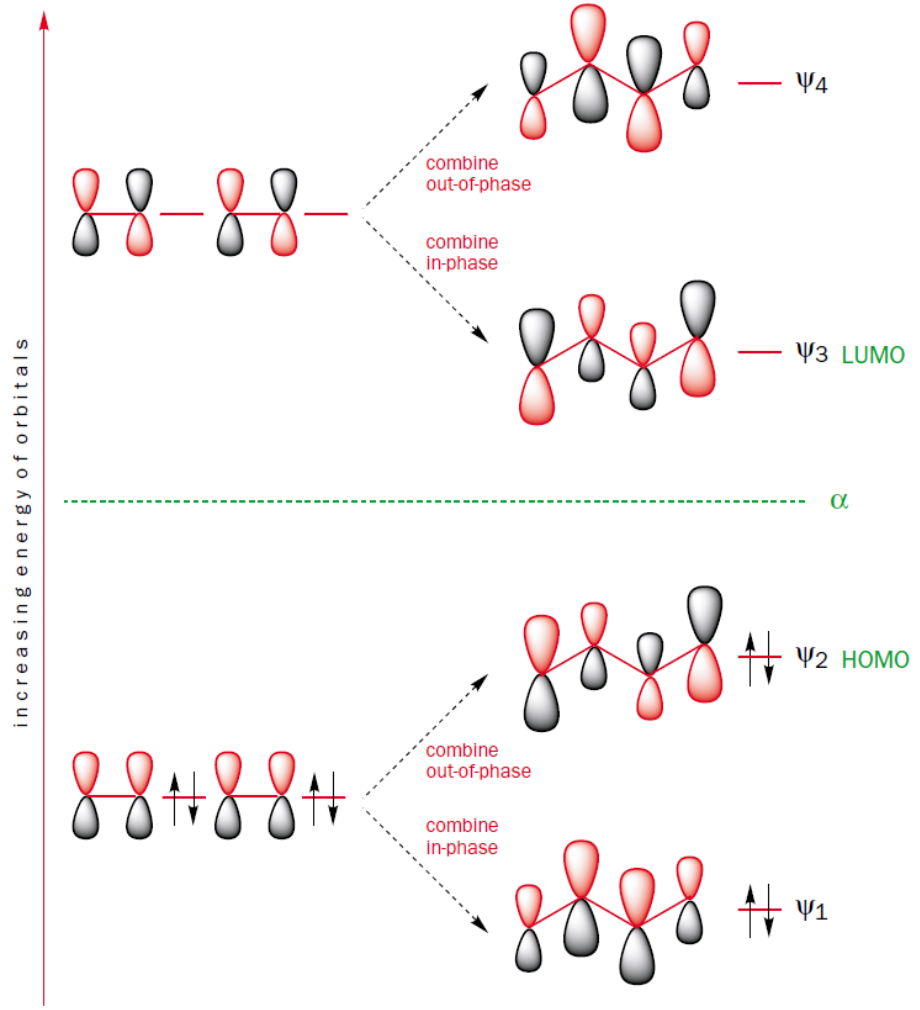


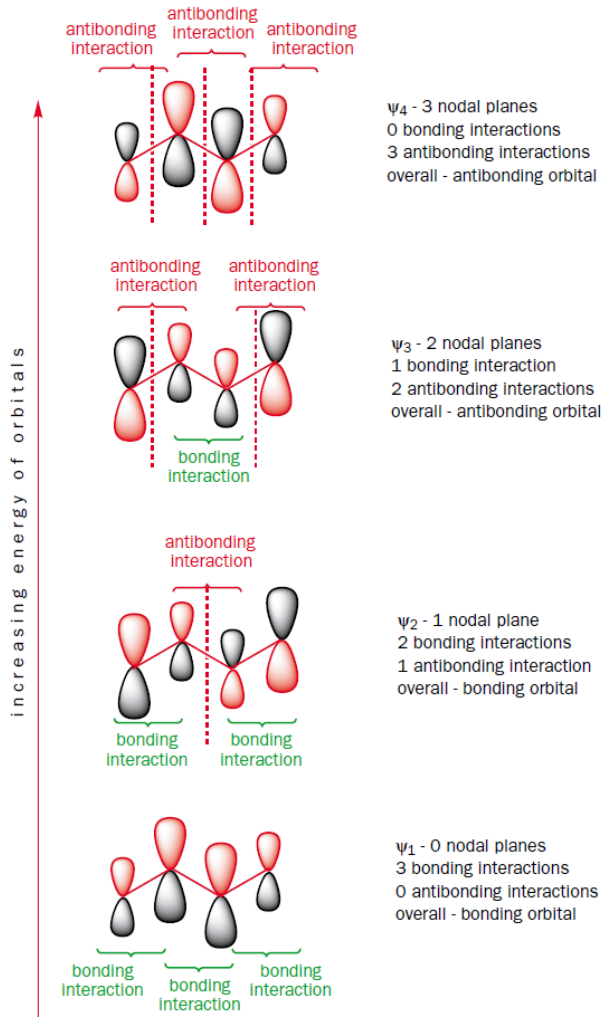
the π bonds formed as a result of the overlap of the individual p orbitals (shown here) must be at right angles to each other



not only are the two π bonds shown in this diagram perpendicular, but the two CH_2 groups are too

1,3-Butadiene orbitals





http://www.chemtube3d.com/orbitalsbutadiene.htm

General Principles for Molecular orbitals arrangement

- **The number of MOs is the same with the number of AOs which combined to produce them**

General Principles for Molecular orbitals arrangement

- **The number of MOs is the same with the number of AOs which combined to produce them**
- **Even number of MOs divide equally in bonding (lower in energy) and anti-bonding (higher in energy)**

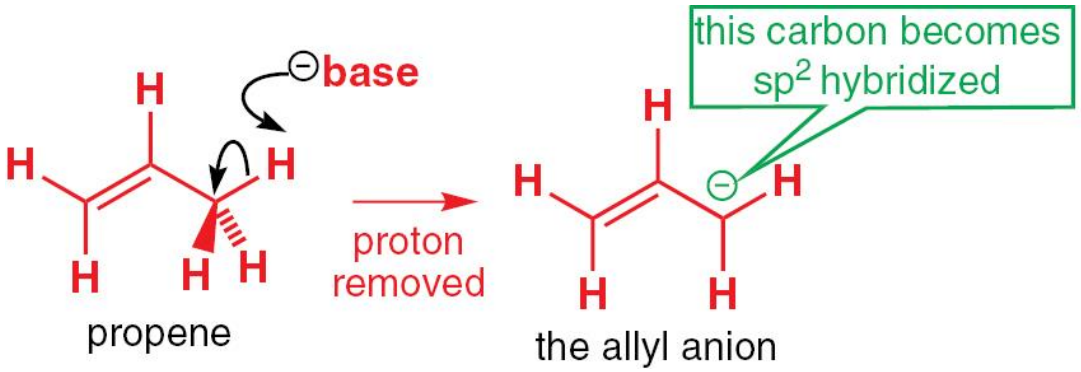
General Principles for Molecular orbitals arrangement

- The number of MOs is the same with the number of AOs which combined to produce them
- Even number of MOs divide equally in bonding (lower in energy) and anti-bonding (higher in energy)
- Odd number of MOs divide equally in bonding and anti-bonding, and an non-bonding MO is found between them

General Principles for Molecular orbitals arrangement

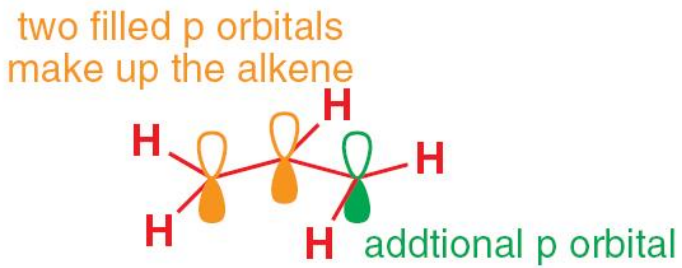
- The number of MOs is the same with the number of AOs which combined to produce them
- Even number of MOs divide equally in bonding (lower in energy) and anti-bonding (higher in energy)
- Odd number of MOs divide equally in bonding and anti-bonding, and an non-bonding MO is found between them
- The nodes are always symmetrically placed and cancel the contribution of any atom through which they pass

Allyl anion- The conjugation of 3 p orbitals



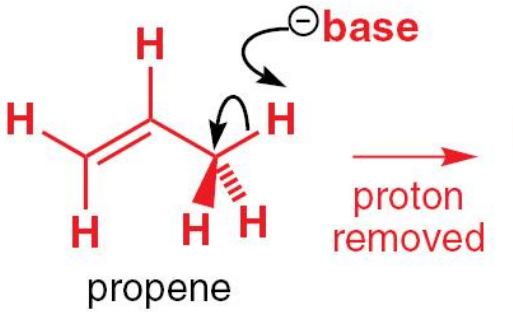
propene

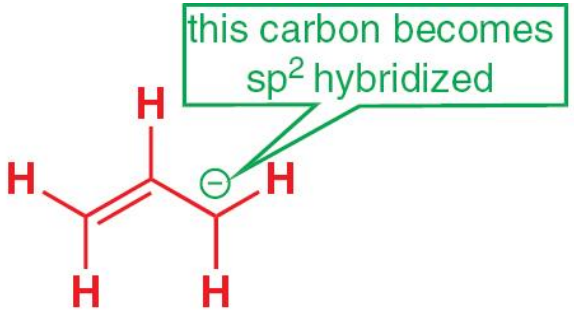
the allyl anion

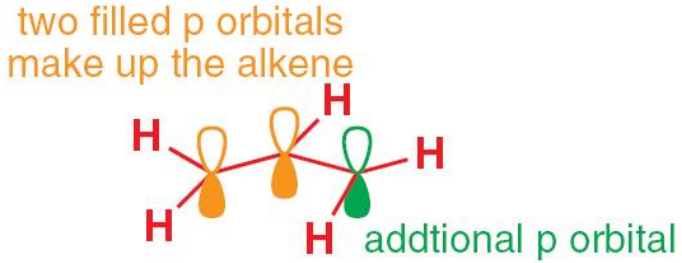


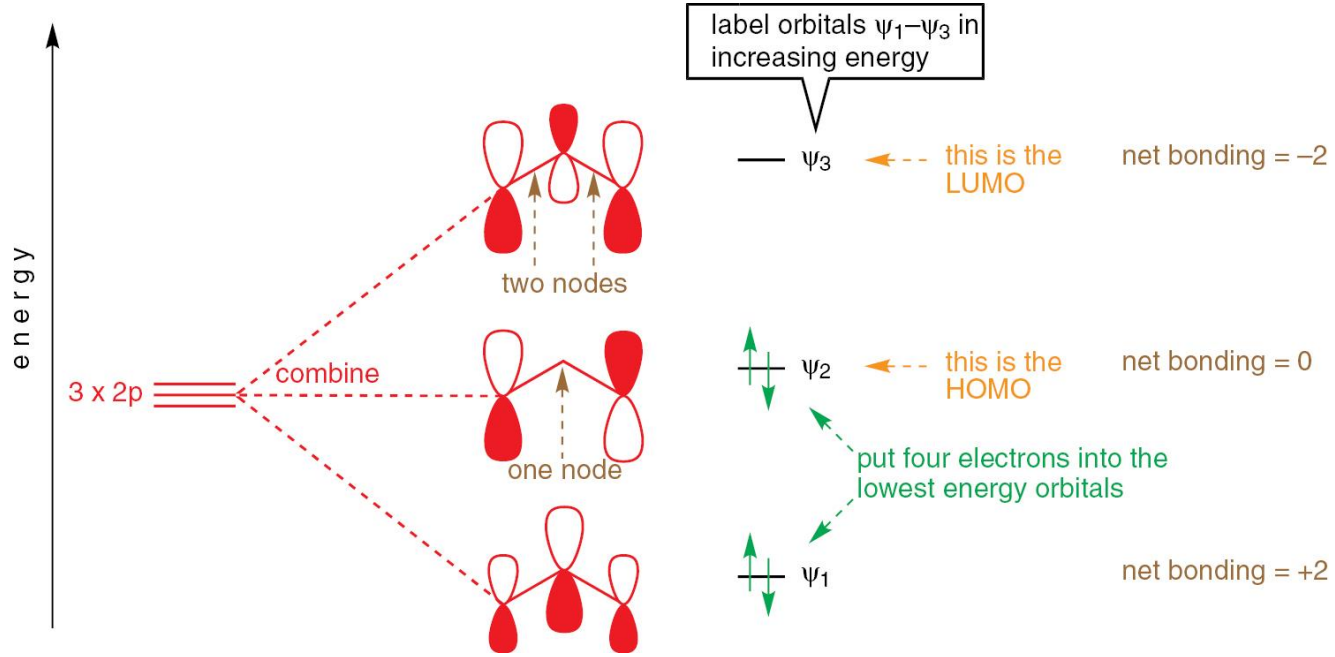
p orbitals of the allyl anion

Allyl anion- The conjugation of 3 p orbitals

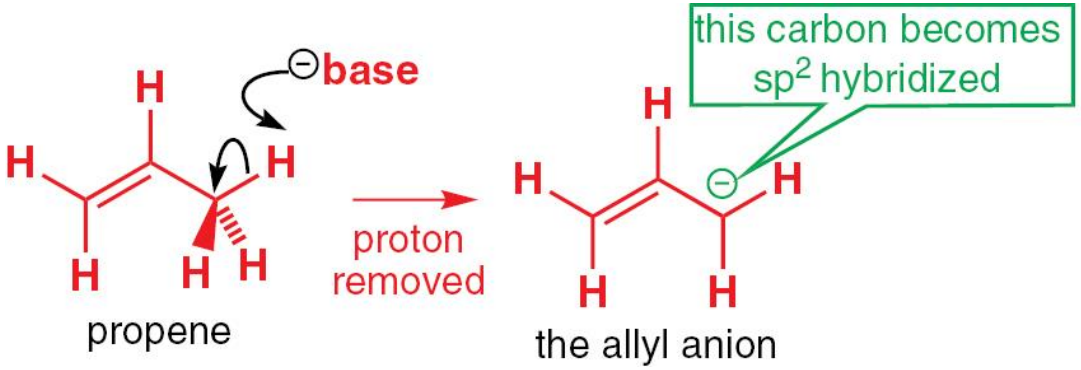


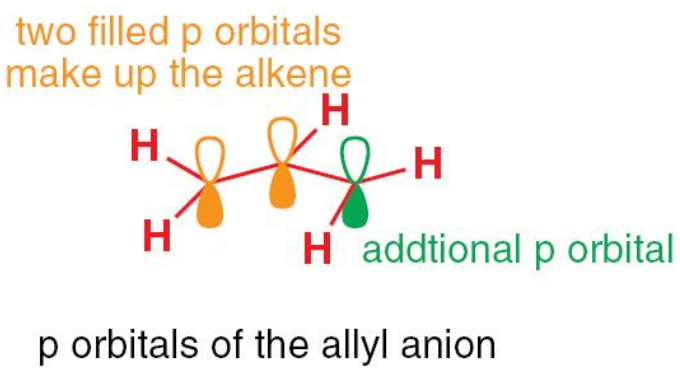




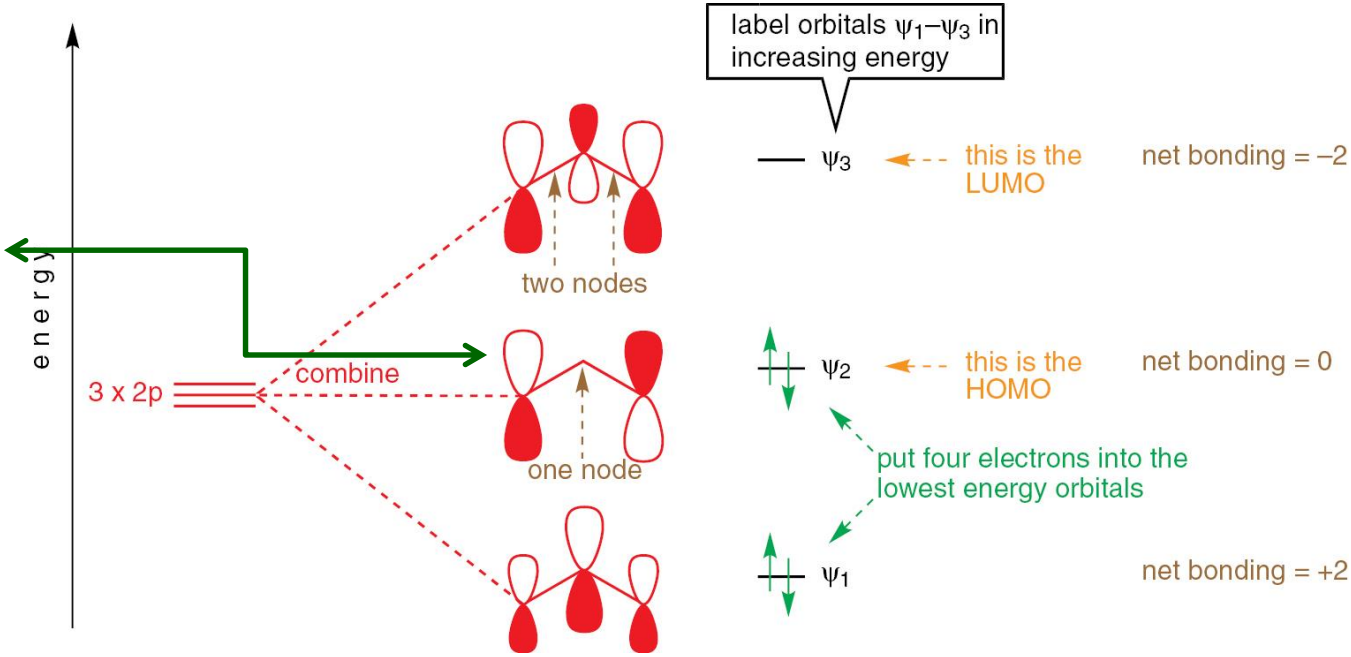


Allyl anion- The conjugation of 3 p orbitals





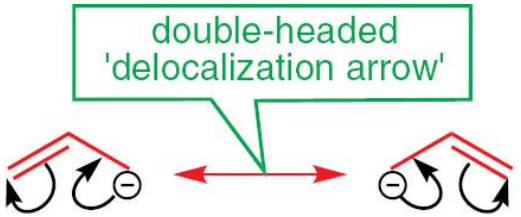
When this MO is occupied there will be no electron density on the central C atom: A non-bonding orbital



Representation of allyl anion



these structures emphasize the equivalence of the bonds and delocalization of charge



the curly arrows show the negative charge concentrated on the end carbon atoms



delocalization arrow

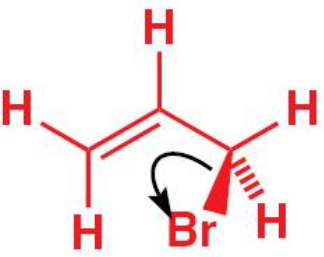


NOT

equilibrium arrow

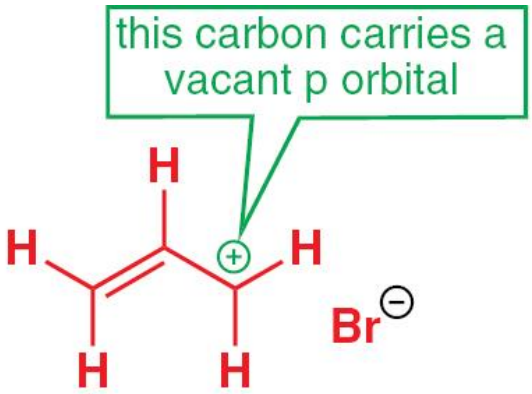


Allyl cation- The conjugation of 3p orbitals

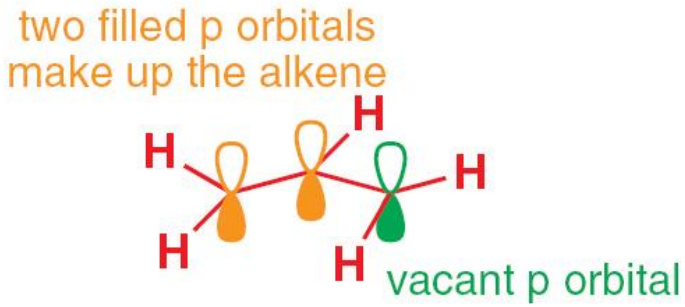


allyl bromide

bromide leaves

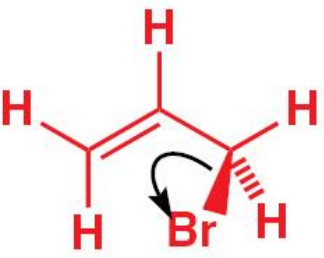


the allyl cation



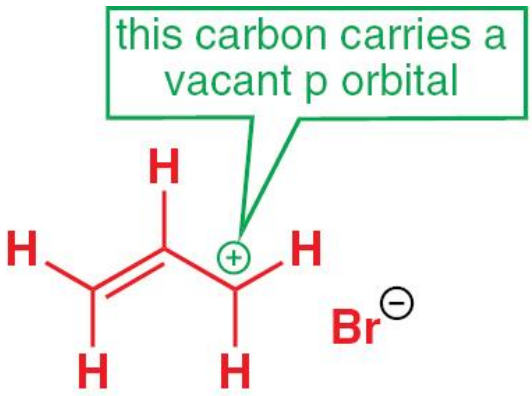
p orbitals of the allyl cation

Allyl cation- The conjugation of 3p orbitals

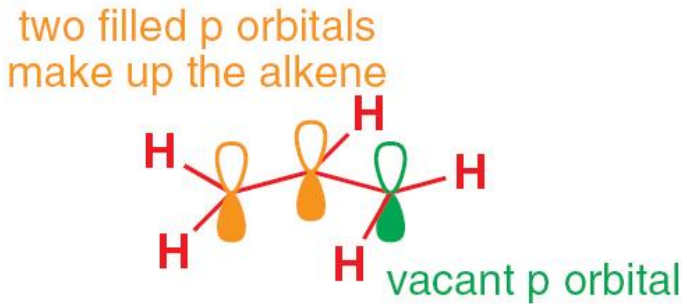


allyl bromide

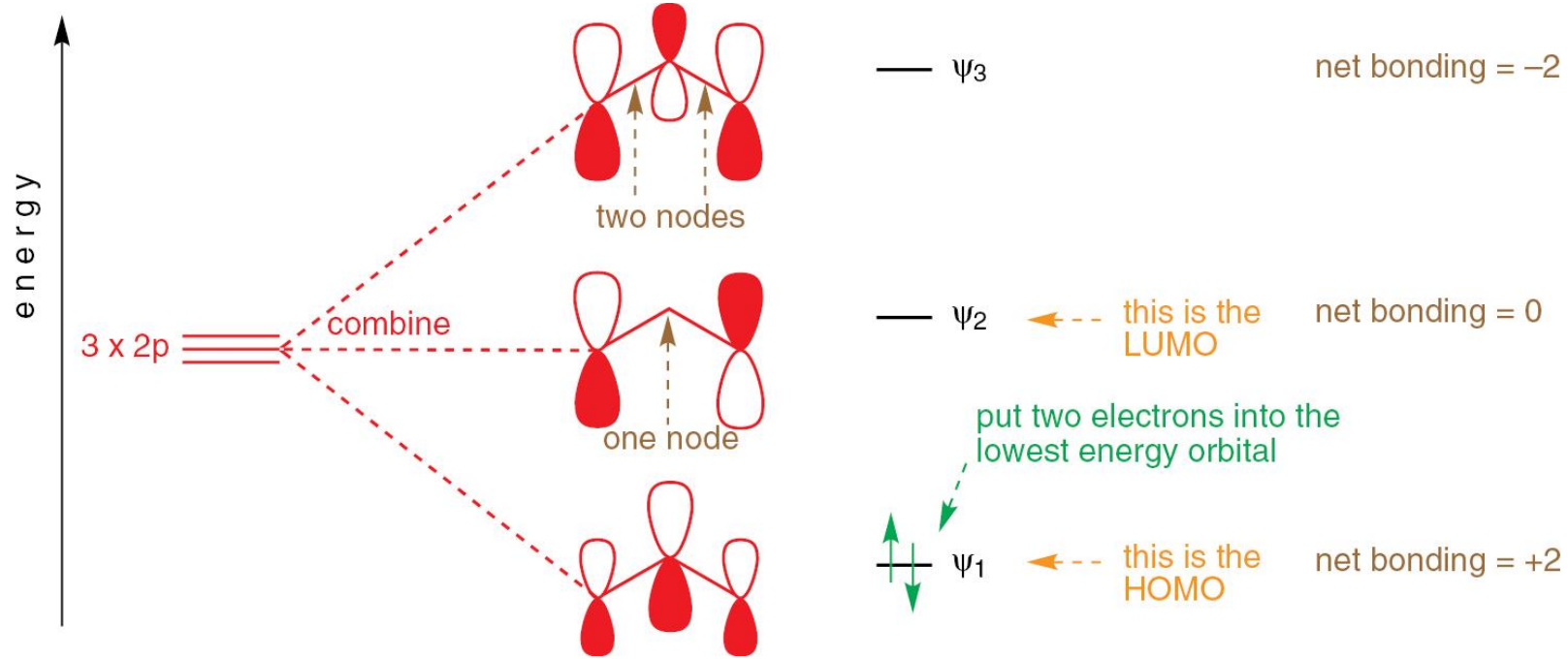
bromide leaves



the allyl cation



p orbitals of the allyl cation



Representation of allyl cation



these structures emphasize the equivalence of the bonds and delocalization of charge



curly arrows show the positive charge is shared over both the end atoms

delocalization arrow

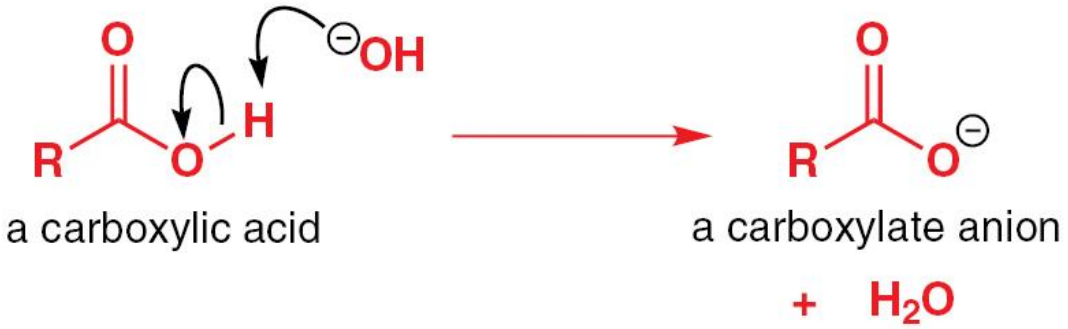


NOT

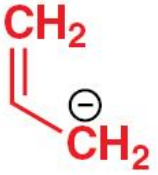
equilibrium arrow



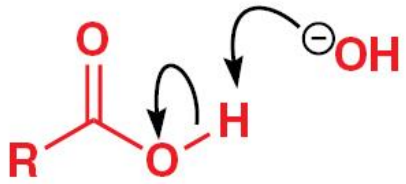
Delocalization over 3 atoms: The carboxylate anion



compare with the allyl anion:



Delocalization over 3 atoms: The carboxylate anion



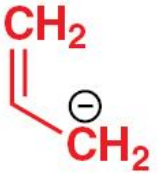
a carboxylic acid



a carboxylate anion

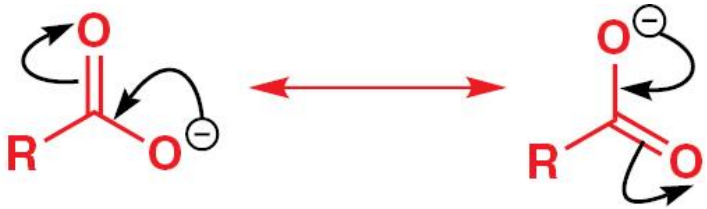
+ H_2O

compare with the allyl anion:



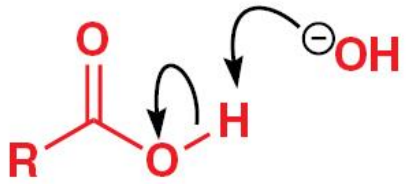


these structures emphasize the equivalence of the two C–O bonds



the electrons are delocalized over the π system

Delocalization over 3 atoms: The carboxylate anion



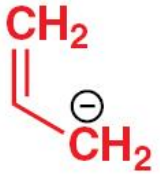
a carboxylic acid



a carboxylate anion

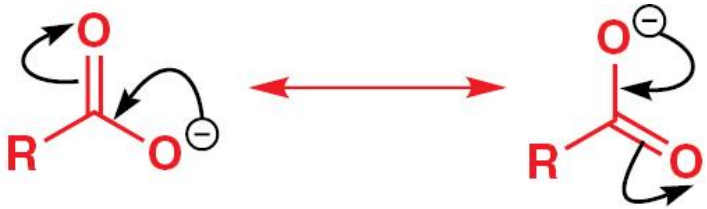
+ H_2O

compare with the allyl anion:





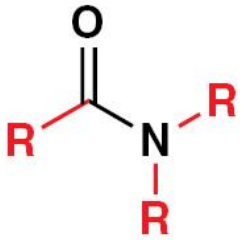
these structures emphasize the equivalence of the two C–O bonds



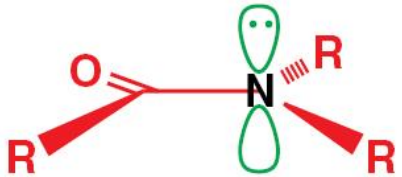
the electrons are delocalized over the π system

Remember the unicorn!

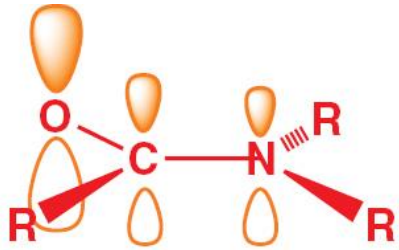
Delocalization over 3 atoms: The amide group



an amide

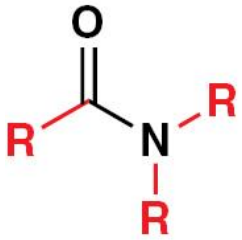


nitrogen is trigonal (sp^2) with its lone pair in a p orbital

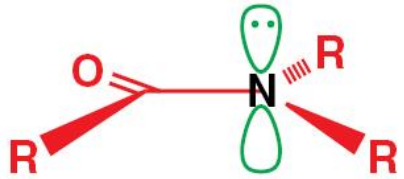


the lowest enegy π orbital of the amide

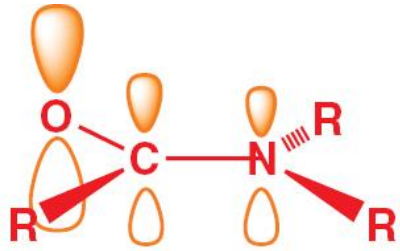
Delocalization over 3 atoms: The amide group



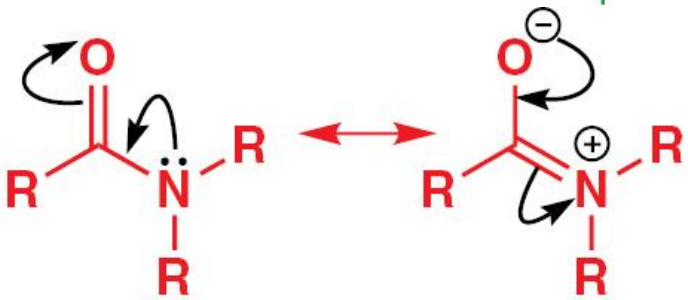
an amide



nitrogen is trigonal (sp^2) with its lone pair in a p orbital

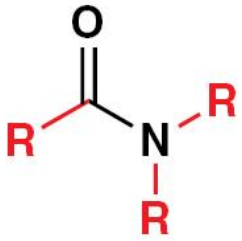


the lowest enegy π orbital of the amide

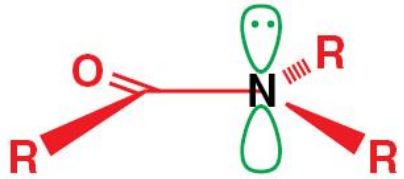


delocalization in the amide group

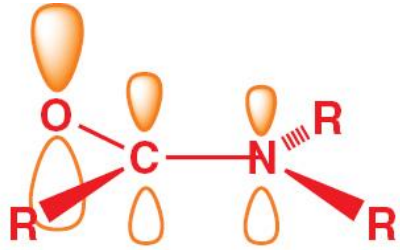
Delocalization over 3 atoms: The amide group



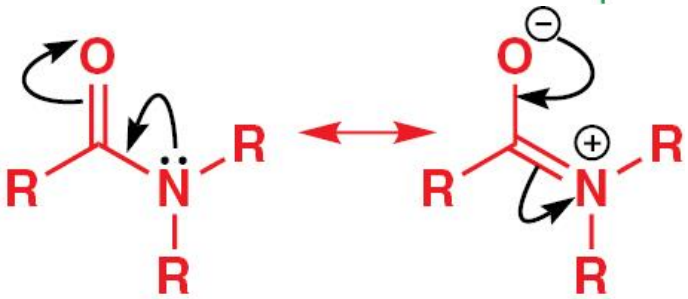
an amide



nitrogen is trigonal (sp^2) with its lone pair in a p orbital



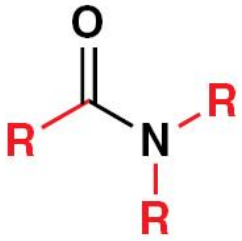
the lowest enegy π orbital of the amide



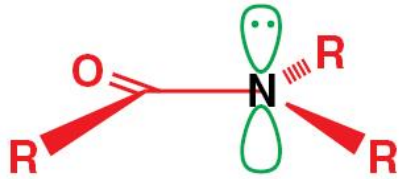
delocalization in the amide group

- **The C-N bond has partial double bond character**
- **The lone pair of N is no longer available**
- **Rotation about the C-N bond is restricted at room temperature**

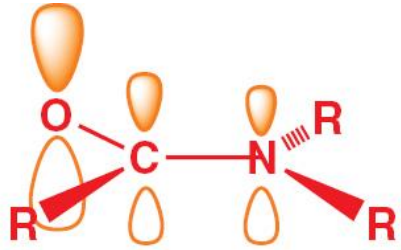
Delocalization over 3 atoms: The amide group



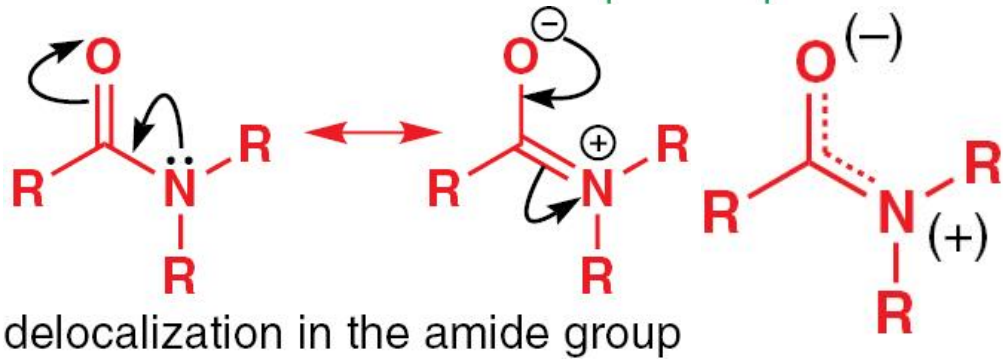
an amide



nitrogen is trigonal (sp^2) with its lone pair in a p orbital



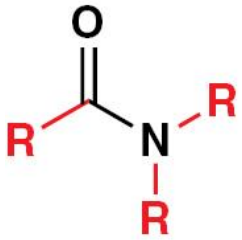
the lowest enegy π orbital of the amide



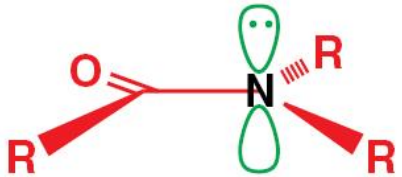
delocalization in the amide group

- **The C-N bond has partial double bond character**
- **The lone pair of N is no longer available**
- **Rotation about the C-N bond is restricted at room temperature**

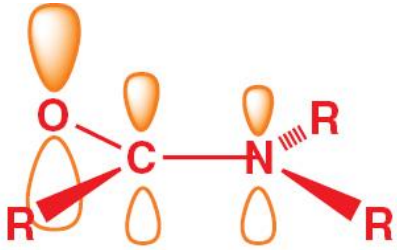
Delocalization over 3 atoms: The amide group



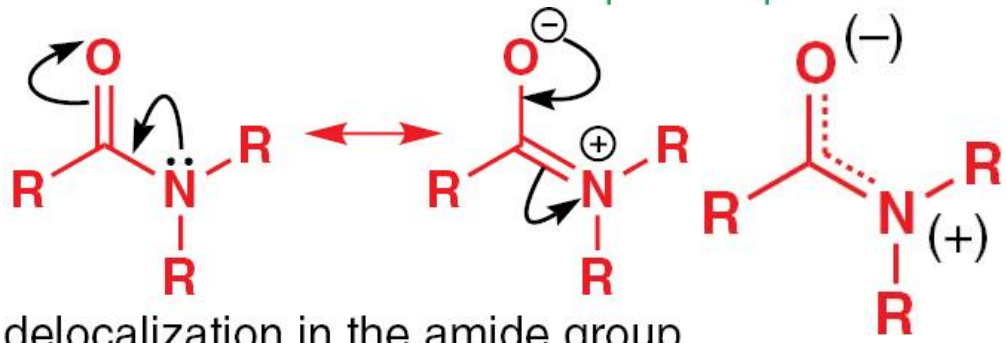
an amide



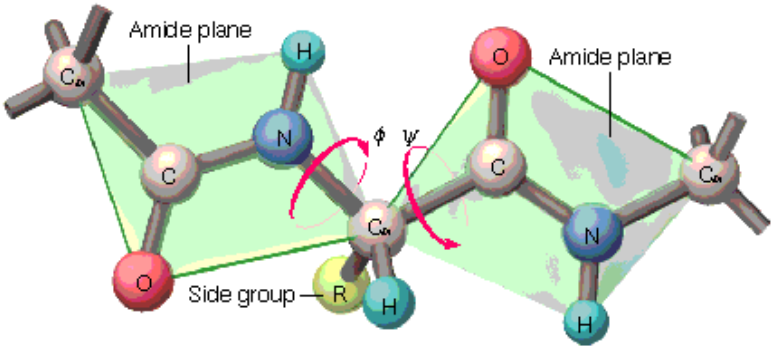
nitrogen is trigonal (sp^2) with its lone pair in a p orbital



the lowest enegy π orbital of the amide



delocalization in the amide group



- **The C-N bond has partial double bond character**
- **The lone pair of N is no longer available**
- **Rotation about the C-N bond is restricted at room temperature**

Delocalization over 3 atoms: The nitro group

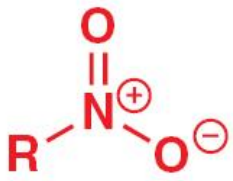


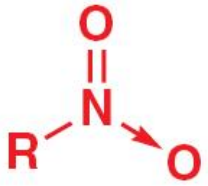
two ways of representing the nitro group



incorrect drawing of the nitro group nitrogen cannot have five bonds

Delocalization over 3 atoms: The nitro group

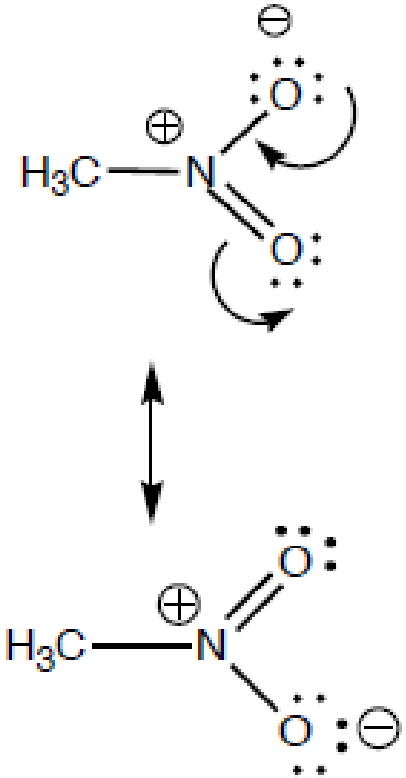




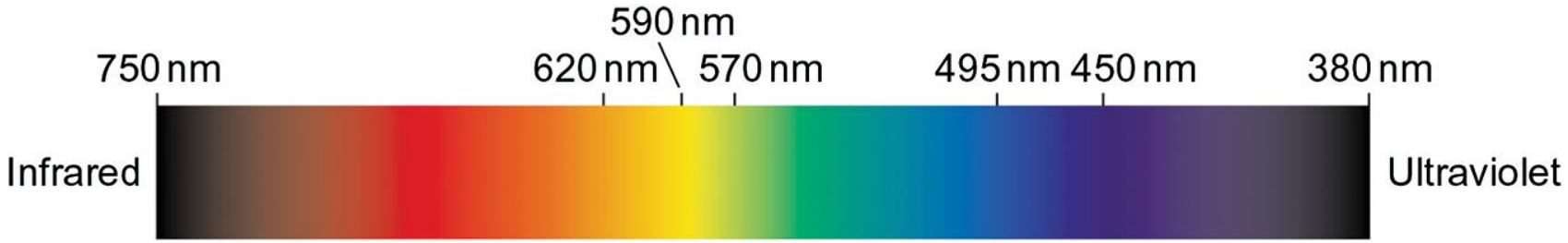
two ways of representing the nitro group



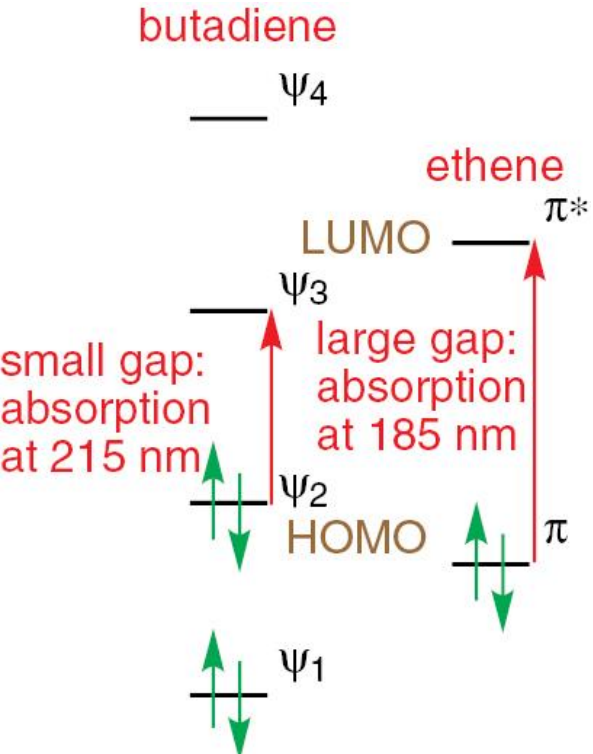
incorrect drawing of the nitro group nitrogen cannot have five bonds



Where does colour come from? UV and Visible Spectra

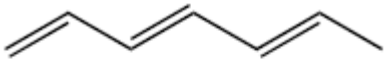


Increasing Energy



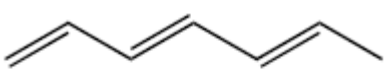
- **Colour comes from transmission of the portion of energy that is not absorbed by an object**
- **Increased degree of conjugation means lower energy needed to move electrons from the ground state to an excited state**

Conjugation and colour

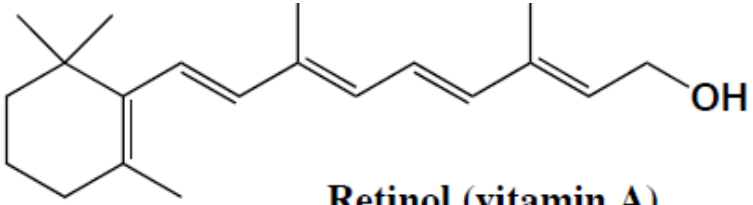


1,3,5-Heptatriene
(3 conjugated C=C bonds)
(colorless)

Conjugation and colour

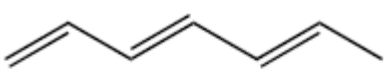


1,3,5-Heptatriene
(3 conjugated C=C bonds)
(colorless)

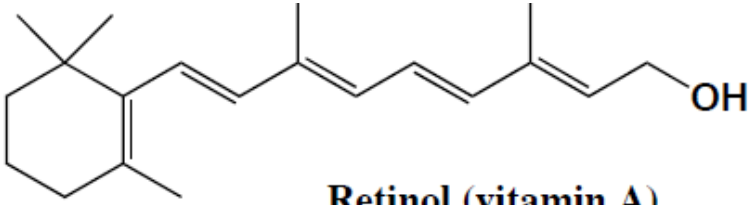


Retinol (vitamin A)
(5 conjugated C=C bonds)
yellow color)

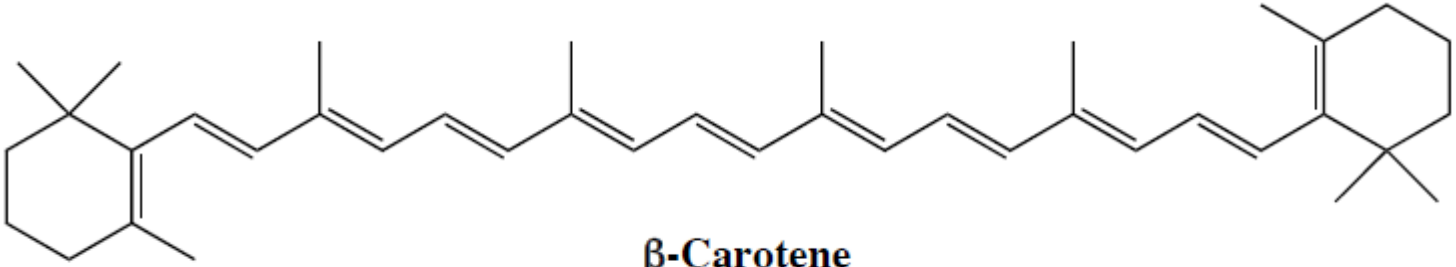
Conjugation and colour



1,3,5-Heptatriene
(3 conjugated C=C bonds)
(colorless)

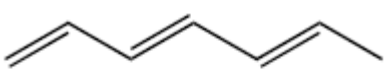


Retinol (vitamin A)
(5 conjugated C=C bonds)
yellow color)

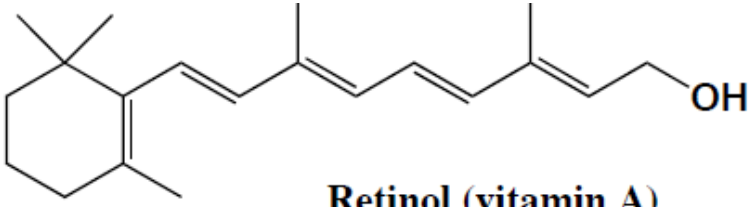


β-Carotene
(11 conjugated C=C bonds)
(orange color)

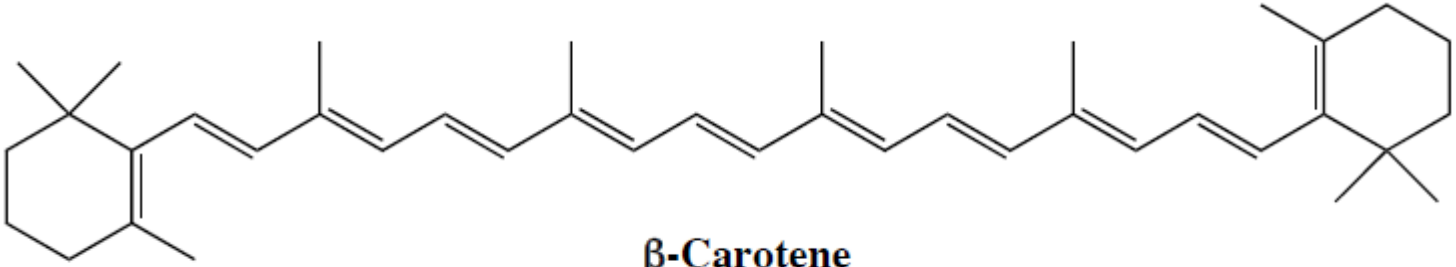
Conjugation and colour



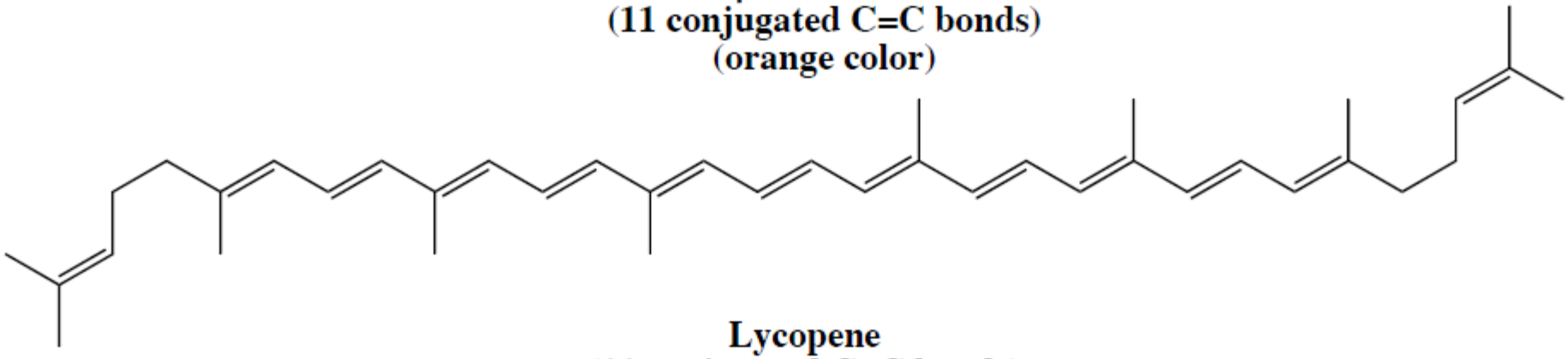
1,3,5-Heptatriene
(3 conjugated C=C bonds)
(colorless)



Retinol (vitamin A)
(5 conjugated C=C bonds)
yellow color)



β-Carotene
(11 conjugated C=C bonds)
(orange color)

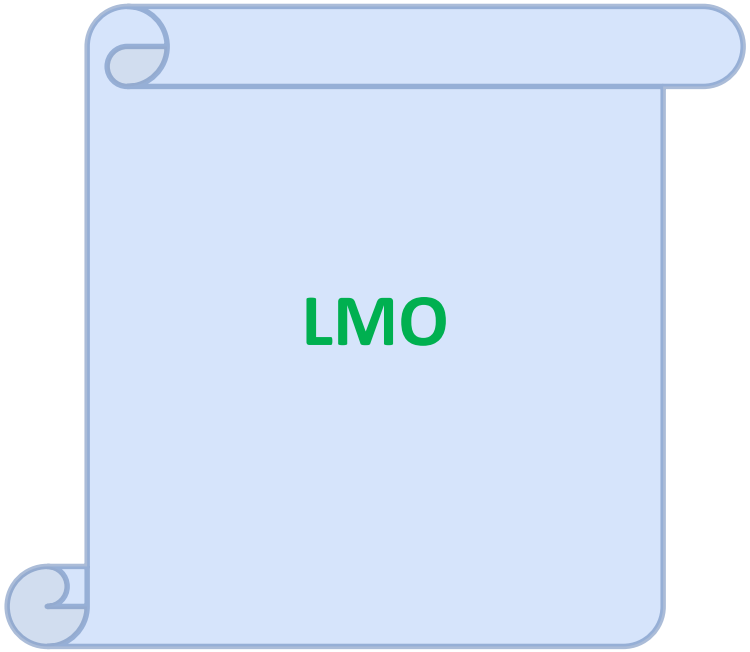


Lycopene
(11 conjugated C=C bonds)
(red color)

Summary

- **Conjugation refers to the sequence of alternating single and double bonds**
- **Delocalization refers to the MOs covering the whole system**
- **Electrons are delocalized over the whole of a conjugated system**
- **Interaction between orbitals over many bonds**
- **Stabilization by the sharing of electrons over more than two atoms**
- **Molecular shape and structure determine reactivity**
- **Representing one aspect of structure by curly arrows**
- **Structure of aromatic compounds**
- **Where colour comes from**

Homework



Study Chapter 7 in textbook!